The
Great Convergence

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CHAPTER GC-1

The Room

On the rooms where the rules were written, the room that never convened, and the question that closes the trilogy

The Room

On May 25, 1787, fifty-five men gathered in a room on the first floor of the Pennsylvania State House in Philadelphia.

The windows were nailed shut — not against the heat, which was considerable, but against eavesdroppers. Armed sentries stood at the doors.

The delegates had been sent by twelve of the thirteen states to revise the Articles of Confederation, the governing document of the young republic. What they did instead was write a new one.

The debates were vicious. Large states fought small states over representation. Slave states fought free states over whether enslaved people would be counted for purposes of congressional apportionment. Delegates from New York walked out in protest and did not return. Luther Martin of Maryland left in disgust. Edmund Randolph of Virginia, who had introduced the Virginia Plan that formed the basis of the Constitution, ultimately refused to sign the finished document because he believed it concentrated too much power in the federal government.

The document they produced was imperfect by any measure.

It sanctioned slavery.

It excluded women from political participation.

It gave no protections to Indigenous nations.

It counted enslaved human beings as three-fifths of a person for the purpose of distributing political power to the states that held them in bondage.

It would require a civil war, twenty-seven amendments, and two and a half centuries of litigation and social movement to begin approaching the ideals stated in its preamble.

It has also governed the most powerful nation on earth for 237 years.

This is what governance looks like when the room convenes. Imperfect. Contested. Stained by the interests of whoever happens to be sitting at the table.

And enduring — because imperfect governance, once written, can be amended, challenged, and improved.

The absence of governance cannot.

> *"The document was imperfect by any measure. It sanctioned slavery. It excluded women. It would require a civil war and twenty-seven amendments. It has also governed the most powerful nation on earth for 237 years. This is what governance looks like when the room convenes."*

The Other Rooms

The Philadelphia convention was not the first room and it was not the last.

The history of governance is the history of rooms — specific physical spaces where, at the critical moment, people gathered and wrote the rules that would shape the world for generations.

In 1944, delegates from forty-four nations gathered at the Mount Washington Hotel in Bretton Woods, New Hampshire.

The world was still at war.

The delegates were there to write the rules for the international financial system that would govern the postwar economy. Over three weeks, they established the International Monetary Fund, the World Bank, and a system of fixed exchange rates anchored to the US dollar.

The framework they produced governed international finance for nearly thirty years and, in modified form, shapes the global financial architecture to this day.

In 1949, diplomats gathered in Geneva to write the rules for the conduct of war.

Two world wars had demonstrated, with industrial precision, what happened when the rules were absent or inadequate: the trenches of the Somme, the firebombing of Dresden, the Holocaust, Hiroshima.

The Geneva Conventions established protections for prisoners of war, wounded combatants, and civilians in conflict zones.

The rules have been violated in every conflict since.

They have also provided the legal framework by which those violations are identified, documented, prosecuted, and — in some cases — punished.

The alternative is not better rules.

The alternative is no framework at all.

In 1968, representatives from 62 nations signed the Treaty on the Non-Proliferation of Nuclear Weapons — the NPT.

The treaty was an imperfect bargain: the five existing nuclear powers agreed to pursue disarmament (a commitment honored more in rhetoric than in practice), and the non-nuclear states agreed not to acquire nuclear weapons (a commitment most have kept).

The NPT has not eliminated nuclear weapons.

It has prevented their proliferation to dozens of nations that had the technical capacity to build them.

The framework, imperfect as it is, has held for nearly sixty years.

No nuclear weapon has been used in war since 1945. Philadelphia. Bretton Woods. Geneva.

The NPT.

In each case, the pattern is the same: a technology or a capability that threatens to produce catastrophic consequences if left ungoverned. A room that convenes. People who argue, compromise, and produce a framework that is imperfect, contested, and amended over time.

And a result that — despite its imperfections — provides the structure within which human beings can hold the technology and its consequences accountable.

The Room That Never Convened

On March 12, 1989, a British computer scientist named Tim Berners-Lee submitted a proposal to his employer, CERN, the European Organization for Nuclear Research.

The document was titled "Information Management: A Proposal." His supervisor wrote on the cover page: "Vague, but exciting." The proposal described a system for linking documents across a network using hypertext — what would become the World Wide Web.

No room convened.

No Philadelphia.

No Geneva.

No Bretton Woods.

The technology that would reshape commerce, communication, politics, journalism, warfare, education, entertainment, and the inner lives of billions of people was deployed globally without a governance framework.

The decisions about how the internet would be structured, who would control its infrastructure, what rules would govern the content that flowed through it, and how the data generated by its users would be collected, stored, sold, and weaponized — these decisions were not made in a room of delegates.

They were made in product meetings, in engineering standups, in venture capital boardrooms, by people optimizing for growth, engagement, and shareholder return.

The result is the world we live in. A world in which five companies control the majority of internet traffic, digital advertising, cloud computing, and online commerce. A world in which a social media platform's recommendation algorithm, optimized for engagement, amplified content that a United Nations fact-finding mission determined played a "determining role" in inciting genocide — a cost documented in *The Last Keeper*. A world in which the surveillance infrastructure described in *The Last Keeper*, Chapter 10, monitors entire populations using the data architecture the internet made possible.

A world in which the concentration of digital power traced across five millennia in *Who Holds the Jar?* has reached its most extreme expression in the platforms that consolidated control of the internet in the absence of governance.

The internet is not ungoverned because governance was tried and failed.

It is ungoverned because the room never convened.

The technology moved faster than the institutions that might have governed it.

By the time regulators understood what the internet had become — GDPR arrived in 2018, thirty years after the web was invented, forty-nine years after ARPANET — the consolidation was complete.

The five companies were entrenched.

The surveillance architecture was built.

The recommendation algorithms were running.

The rules, such as they were, had already been written by the companies themselves, for themselves.

> *"The internet is not ungoverned because governance was tried and failed. It is ungoverned because the room never convened. The technology moved faster than the institutions that might have governed it."*

The Pattern

There is a pattern in this history, and naming it is the purpose of this book.

Every transformative technology passes through a window — a finite period of time during which the rules that will govern the technology for the next generation, or the next century, are written.

The window opens when the technology is powerful enough to produce consequences that demand governance but not yet so entrenched that governance is impossible.

The window closes when the technology's consolidation becomes irreversible — when the infrastructure is built, the dependencies are established, and the power dynamics are locked in. During the window, there is a choice.

The room can convene. People can argue, compromise, and produce a framework — imperfect, as all governance is, but real. Or the room can fail to convene, and the vacuum will be filled by the entities that consolidated control of the technology.

The rules will still be written — by the consolidators, for the consolidators, embedded in terms of service and algorithmic design choices and corporate policies that function as governance without the legitimacy, the accountability, or the democratic mandate that governance requires.

The printing press. Electricity. Nuclear energy.

The internet.

Each technology reached its governance window. Some were governed — imperfectly, belatedly, but governed.

One was not.

The cost of the one that was not is documented in two books that precede this one: power concentrated without check, and humans removed from the loops that mattered most, with consequences measured in lives.

The Question

This book is about the governance window for artificial intelligence.

It is the third and final book in a trilogy called *The Cost of the Machine*.

The first book — *The Last Keeper* — documented the cost: what happens when humans are removed from the loops that matter most. Patients irradiated by a machine whose safety interlock was replaced by software. Pilots who could not fly the aircraft the autopilot had been flying for them. Defendants sentenced by an algorithm they could not cross-examine. A recommendation engine that optimized for engagement and produced genocide. Targeting systems that generate kill lists faster than humans can review them.

The cost is measured in lives, liberty, and justice.

The second book — *Who Holds the Jar?* — documented the pattern: computational power has concentrated in fewer hands with every era, from the shepherd's counting jar in ancient Mesopotamia through the mainframe, the personal computer, the internet, and now artificial intelligence.

Each era distributes capability and then recaptures it.

The concentration is not accidental.

It is structural — driven by the economics of infrastructure, the advantages of scale, and the absence of governance that might distribute power differently.

This book asks the question that follows from both: who writes the rules?

And what happens when nobody does?

The governance window for AI is open.

The innovation phase is complete — the breakthroughs in deep learning between 2012 and 2020 produced the capability.

The democratization phase is accelerating — ChatGPT, open-source models, AI code generation tools have placed the technology in the hands of hundreds of millions of people.

The chaos phase is beginning — disinformation, labor displacement, ungoverned deployment in healthcare, criminal justice, military operations, and critical infrastructure.

The consolidation phase is already visible — seven organizations capable of training frontier models, two chip manufacturers, five cloud providers, one orbital communications constellation that controls more than sixty percent of active satellites.

The room has not yet convened. Fragments of governance exist — the EU AI Act, executive orders, national strategies, corporate self-regulation.

But no Philadelphia.

No Bretton Woods.

No NPT for artificial intelligence.

No single framework that addresses the technology's global reach with governance of equivalent scope.

The window is open.

It will not stay open. History says it closes when the consolidation becomes irreversible.

The question this book asks — the question the trilogy ends on — is whether the room will convene before it does.

> *"The window is open. It will not stay open. History says it closes when the consolidation becomes irreversible. The question is whether the room will convene before the window shuts."*

What This Book Does

This book traces the governance pattern across the full historical arc — from the printing press to artificial intelligence — through a five-phase cycle that every transformative technology has followed: Innovation, Democratization, Chaos, Consolidation, and Governance.

The cycle is not a metaphor.

It is empirically observable across the documented cases.

Part I establishes the pattern through history.

Part II documents the chaos phase of AI — what is happening right now, to real people, in software, in information, and in labor.

Part III maps the consolidation — who is gaining control and what they are doing with it.

Part IV enters the room — who is writing the rules, who is absent, and what the convergence of power, cost, and governance looks like at the moment of decision.

Part V makes the closing argument of the trilogy.

The question is not whether AI should be governed.

Every powerful technology has been governed, sooner or later, well or badly.

The question is whether the governance will arrive during the window — proactively, shaped by the people who will live under it — or after the window closes, reactively, shaped by the catastrophes that forced the response.

The printing press was governed after the Wars of Religion. Nuclear energy was governed after Hiroshima. Aviation was governed after enough people died.

The cost of governing after the catastrophe is the cost documented in *The Last Keeper*.

The pattern that produces the catastrophe is the pattern documented in *Who Holds the Jar?*.

And the choice — whether to convene the room before the window closes — is the subject of this book.

* * *

CHAPTER GC-2

The Cycle

On the five-phase pattern that every transformative technology has followed, from the Gutenberg press through electricity through nuclear energy, and the discovery that the governance window is shrinking with each era

The Cycle

In 1440, a goldsmith in Mainz named Johannes Gutenberg began experimenting with a device that would become the most consequential invention of the second millennium. His printing press with movable type did not create a new capability — the Chinese had developed movable type centuries earlier, and hand-copying and woodblock printing had existed for thousands of years. What Gutenberg created was a capability that could scale. A single press could produce hundreds of copies of a text in the time it took a scribe to produce one.

The cost of producing a book dropped by approximately eighty percent within fifty years.

The five-phase cycle that the previous chapter named — Innovation, Democratization, Chaos, Consolidation, Governance — through three transformative technologies.

The printing press. Electricity.

And nuclear energy.

Each case demonstrates the same pattern.

Each case shows the governance window shrinking.

And each case reveals a truth that the final section of this chapter will apply to artificial intelligence: the cost of the chaos phase is determined by how long it takes the room to convene.

The Press: 350 Years

Phase 1 — Innovation. Gutenberg's workshop in Mainz. A single inventor, a single press, a single city.

The first major work printed was the Gutenberg Bible, completed around 1455. Approximately 180 copies were produced — a revolutionary number for the era but a rounding error by later standards.

The technology was expensive, fragile, and concentrated in the hands of a small number of skilled operators. Power was concentrated because the technology was scarce.

Phase 2 — Democratization.

By 1500, printing presses had been established in more than 250 cities across Europe. An estimated twenty million volumes had been printed in the first fifty years of the technology's existence.

The price of a book dropped to a fraction of what a hand-copied manuscript had cost. Literacy, which had been confined to clergy and aristocracy, began its centuries-long expansion into the general population.

The technology was no longer scarce.

The capability was distributed.

Phase 3 — Chaos.

This is where the cost was paid.

On October 31, 1517, Martin Luther posted his Ninety-Five Theses on the door of the Castle Church in Wittenberg.

The document was a theological critique of the Catholic Church's sale of indulgences.

Within weeks, printed copies had spread across Germany.

Within months, translations had reached every corner of Europe. Luther's ideas, which in the pre-print era might have remained the concern of a small circle of academics, ignited the Protestant Reformation — the most consequential schism in the history of Western Christianity. What followed was approximately 130 years of religious warfare.

The French Wars of Religion (1562–1598) killed an estimated three million people.

The Thirty Years' War (1618–1648), which engulfed most of continental Europe, killed an estimated eight million — roughly twenty percent of the German population.

The English Civil War (1642–1651) killed approximately 200,000 in a nation of five million.

The printing press did not cause these wars. Religious tensions, political rivalries, and territorial ambitions would have produced conflict regardless.

But the press *amplified* and *accelerated* the conflict by making it possible, for the first time in history, to distribute ideas — including inflammatory ideas, including calls to violence, including propaganda designed to dehumanize the enemy — at a speed and scale that existing governance structures could not contain.

The printing press was, in the terms *The Last Keeper* documented, a recommendation engine in the sixteenth century.

The technology amplifies whatever produces the strongest reaction.

In Myanmar in 2017, the strongest reaction was anti-Rohingya violence.

In Europe in the sixteenth century, the strongest reaction was religious hatred.

> *“The printing press did not cause the wars. It amplified them — distributing ideas at a speed that existing governance could not contain. The printing press was a recommendation engine in the sixteenth century.”*

Phase 4 — Consolidation. Governments responded not with governance but with control.

The Catholic Church published the Index Librorum Prohibitorum — the list of prohibited books — in 1559.

The English crown imposed licensing requirements on printers through the Licensing of the Press Act of 1662, restricting printing to approved operators. State censorship regimes arose across Europe, concentrating control of the printing press in the hands of the governments and institutions that had the power to regulate it.

The technology had been distributed.

The power over it was recaptured.

Phase 5 — Governance. True governance — frameworks that balanced the technology’s power against its risks while preserving its benefits — arrived slowly.

The Statute of Anne, passed by the English Parliament in 1710, established the first modern copyright law, creating a legal framework for who could reproduce printed works and under what conditions.

The First Amendment to the US Constitution, ratified in 1791, established press freedom as a protected right — governance not of the press itself but of the government’s power to suppress it. Libel and defamation law developed over centuries to address the press’s capacity for harm. From Gutenberg’s press to the First Amendment: approximately 350 years. From Innovation to Governance: three and a half centuries.

And the Chaos phase — the period between democratization and the first meaningful governance — lasted roughly 130 years and cost millions of lives.

Electricity: 50 Years

Phase 1 — Innovation.

On September 4, 1882, Thomas Edison flipped a switch at the Pearl Street Station in lower Manhattan. Electricity flowed to 85 customers within a one-square-mile radius. Among the first was J.P. Morgan, whose office at 23 Wall Street had been wired in advance.

The technology was revolutionary, expensive, and available only to the wealthy and the connected.

Phase 2 — Democratization. Within two decades, electrical grids expanded across American and European cities. Nikola Tesla's alternating current system, championed by George Westinghouse, proved that electricity could be transmitted over long distances — enabling the electrification of regions far from generating stations. Factories converted from steam to electric power. Electric lighting replaced gas. Streetcars electrified urban transportation.

By 1920, approximately 35% of American homes had electricity.

The technology was no longer a luxury for J.P. Morgan.

It was becoming infrastructure.

Phase 3 — Chaos.

The electrification of industry produced consequences that no governance framework was prepared to address. Factories that had relied on human and steam power reorganized around electric motors, increasing productivity and eliminating jobs. Workers who resisted the changes were replaced. Workers who adapted faced new dangers: electrocution, industrial accidents in electrically powered machinery, and the relentless acceleration of production speed that electric power made possible.

The electrical industry itself consolidated rapidly. Edison, Westinghouse, and their competitors fought the "War of Currents" — a commercial battle over whether direct current (Edison) or alternating current (Westinghouse/Tesla) would become the standard.

The war was fought not just in the marketplace but in the press and in public demonstrations designed to frighten consumers about the rival technology. Edison publicly electrocuted animals to demonstrate the dangers of alternating current.

The first electric chair — using alternating current — was promoted by Edison's allies as evidence that Westinghouse's technology was suitable for killing. Meanwhile, the companies that controlled electrical generation and distribution became monopolies. Samuel Insull, who had started as Edison's personal secretary, built a utility empire that controlled electricity for millions of customers across the Midwest. His holding company structure — companies owning companies owning companies, each layer adding leverage and obscuring risk — became the template for the utility industry.

When the structure collapsed during the Great Depression, investors lost hundreds of millions of dollars. Insull fled the country. Customers who depended on his grid were left with infrastructure owned by a bankrupt empire.

> *"Edison publicly electrocuted animals to demonstrate the dangers of his rival's technology. Insull built a utility empire on leverage and opacity. When it collapsed, millions of customers depended on infrastructure owned by a bankrupt empire. The chaos phase is never theoretical."*

Phase 4 — Consolidation.

By the 1920s, a small number of utility holding companies controlled the majority of American electrical generation and distribution.

The customers who benefited from cheap electricity were dependent on monopolies they had no power to regulate. The dynamic mirrors what followed the printing press: the technology distributes capability, the market concentrates control. The dynamic holds.

Phase 5 — Governance.

The governance arrived in stages.

The Public Utility Holding Company Act of 1935, passed in response to the Insull collapse and similar scandals, broke up the utility holding companies and established federal regulation of interstate electricity sales.

The Rural Electrification Act of 1936 brought electricity to rural America, where private utilities had refused to invest because the return was insufficient. State public utility commissions established rate-setting authority, ensuring that monopoly providers could not charge whatever the market would bear. From Pearl Street Station to the Public Utility Holding Company Act: approximately 53 years. From Innovation to Governance: half a century.

The chaos phase was shorter than the printing press — decades rather than centuries — but the pattern was identical: innovation, democratization, chaos, consolidation, governance.

And the governance arrived only after the consolidation produced a crisis that forced the political system to respond.

Nuclear Energy: 26 Years

Phase 1 — Innovation.

On July 16, 1945, the first nuclear weapon was detonated at the Trinity test site in New Mexico.

The Manhattan Project had spent approximately $2 billion (roughly $30 billion in today’s dollars) and employed over 125,000 people.

The technology was the most expensive and the most concentrated innovation in human history to that point.

Three weeks later, nuclear weapons were used against the civilian populations of Hiroshima and Nagasaki. Approximately 200,000 people were killed, most of them instantly or within months.

Phase 2 — Democratization.

The Soviet Union tested its first nuclear weapon in 1949 — four years after Hiroshima.

The United Kingdom tested in 1952. France in 1960. China in 1964.

The technology that had been the monopoly of one nation distributed to five within twenty years.

The knowledge spread further: by the 1960s, dozens of nations had the technical capacity to build nuclear weapons if they chose to.

Phase 3 — Chaos.

The chaos phase of nuclear technology was the Cold War itself.

The Cuban Missile Crisis of October 1962 brought the world to the brink of nuclear war — thirteen days during which the continued existence of human civilization depended on the judgment of a small number of individuals operating under extreme pressure with incomplete information.

The doctrine of mutually assured destruction — the principle that any nuclear attack would be met with a retaliatory strike of sufficient magnitude to destroy the attacker — was not a governance framework.

It was the absence of one: a recognition that no rules existed and that the only restraint was the fear of annihilation. During this period, nuclear testing contaminated vast areas of the Pacific, Central Asia, and the American Southwest. Atmospheric testing deposited radioactive fallout across the globe.

The population living downwind of test sites suffered elevated rates of cancer and birth defects for generations.

The chaos was not a war.

It was the slow, distributed poisoning of the planet by a technology that no governance framework constrained.

Phase 4 — Consolidation.

The five nuclear powers consolidated their advantage.

The technology that had been distributed through espionage and parallel development was now something the existing nuclear states sought to prevent from spreading further.

The consolidation was not market-driven, as it had been with the printing press and electricity.

It was strategic: the nations that held the technology wanted to ensure that no additional nations acquired it, because each new nuclear state increased the risk of the catastrophe that no one could control.

Phase 5 — Governance.

The Treaty on the Non-Proliferation of Nuclear Weapons was signed in 1968 and entered into force in 1970.

The treaty was, as noted in the previous chapter, an imperfect bargain.

The nuclear states agreed to pursue disarmament.

The non-nuclear states agreed not to acquire weapons.

The disarmament commitment has been honored more in rhetoric than in action.

But the nonproliferation commitment has largely held: of the dozens of nations that had the technical capacity to build nuclear weapons, only four additional states have done so outside the treaty framework (India, Pakistan, Israel, and North Korea).

The framework is imperfect.

It has also prevented the scenario that the chaos phase seemed to promise: a world with dozens of nuclear-armed states. From Trinity to the NPT: 23 years. From Innovation to Governance: less than a quarter-century.

The governance window was shorter than for electricity, which was shorter than for the printing press.

And the governance was more proactive than in any previous case — driven not by a catastrophe that had already occurred (though Hiroshima and Nagasaki provided the initial impetus) but by the recognition that the catastrophe that *could* occur — global nuclear war — was of sufficient magnitude that waiting for it to happen before writing the rules was an unacceptable risk.

> *"From Gutenberg to the First Amendment: 350 years. From Pearl Street to utility regulation: 53 years. From Trinity to the NPT: 23 years. The window is shrinking. The cost of delay is measured in the chaos phase."*

The Acceleration

Three technologies.

Three instances of the same five-phase cycle.

And one unmistakable trend: the time between innovation and governance is compressing.

The printing press: approximately 350 years from Gutenberg to the Statute of Anne.

The chaos phase lasted roughly 130 years and killed millions. Electricity: approximately 53 years from Pearl Street to the Public Utility Holding Company Act.

The chaos phase lasted roughly 30 years and produced monopoly abuse, industrial devastation, and financial collapse. Nuclear energy: approximately 23 years from Trinity to the NPT.

The chaos phase lasted roughly 17 years and brought the world to the brink of annihilation.

The compression is not accidental.

Each successive technology has been more powerful, more rapidly deployable, and more consequential than the last.

The printing press reshaped information; it took decades to reach a significant fraction of the European population. Electricity reshaped industry and daily life; it took years to reach millions of homes. Nuclear weapons reshaped geopolitics; the capability to destroy civilization was achieved in months.

The consequence of the compression is this: the governance window is not just shorter.

It is shorter while the technology's impact is larger and the complexity of governance is greater.

The room must convene faster, address more consequences, and produce a framework for a technology that is still evolving while the rules are being written.

The printing press was a static technology by the time copyright law arrived. Electricity was well understood by the time utility regulation was enacted. Nuclear weapons were a known quantity by the time the NPT was signed. AI is none of these things.

It is evolving while the room deliberates.

The Cost of the Chaos Phase

The pattern teaches one lesson above all others: the cost of a transformative technology is determined not by the technology itself but by the duration of the chaos phase — the period between the technology's democratization and the arrival of governance.

The printing press was not inherently destructive.

It produced the Reformation, the Enlightenment, modern science, mass literacy, and the democratic revolutions that shaped the modern world.

It also produced 130 years of religious warfare.

The difference between the two outcomes was not the technology.

It was whether governance — rules for how the technology's power would be channeled, constrained, and directed — was present or absent. Electricity was not inherently exploitative.

It produced modern medicine, universal lighting, industrial productivity, and the infrastructure of the twentieth century.

It also produced monopoly abuse, worker displacement, and financial fraud.

The difference was governance. Nuclear energy was not inherently apocalyptic.

It has produced clean electricity, medical imaging, and scientific capabilities that have extended human understanding of the universe.

It also brought the world to the edge of extinction.

The difference was governance. AI is not inherently anything.

It will produce extraordinary benefits — in medicine, in science, in accessibility, in creativity, in the democratization of capability that has been the province of the privileged. It will also produce severe harms — the harms documented across eleven chapters of *The Last Keeper*, the concentration of power documented across fourteen chapters of *Who Holds the Jar?*.

The difference between the two outcomes is the same difference it has always been: governance.

The room must convene.

The rules must be written.

The window is open.

And the history traced in this chapter provides no reason for confidence that the window will stay open for long.

> *“AI is not inherently anything. It will produce extraordinary benefits and severe harms. The difference between the two outcomes is the same difference it has always been: governance. The room must convene. The window is open. It will not stay open for long.”*

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CHAPTER GC-3

The Internet's Lesson

On the technology that was never adequately governed, and the warning it contains for everything that follows

The Internet's Lesson

In 1995, an astronomer named Clifford Stoll published a column in *Newsweek* titled "The Internet? Bah!" He argued that the internet was overhyped, that online databases would never replace daily newspapers, that no online bookstore could match the experience of browsing a physical shop, and that the prediction of a vibrant online community was nonsense. "The truth is," he wrote, "no online database will replace your daily newspaper." Stoll was spectacularly wrong about the internet's potential.

But embedded in his skepticism was a question that proved more important than any of his predictions: who would govern this thing? Who would write the rules for a technology that was, in 1995, already connecting millions of people across borders, across jurisdictions, and across every existing regulatory framework? Stoll did not ask this question. Neither did anyone else — not with the urgency it required.

And that failure to ask is the subject of this chapter.

The Five Phases of the Internet

Phase 1 — Innovation.

The internet's origin is typically dated to October 29, 1969, when the first message was sent over ARPANET, a network funded by the US Department of Defense's Advanced Research Projects Agency.

The message was supposed to be "LOGIN." The system crashed after the first two letters.

The first message transmitted over the network that would become the internet was "LO." For its first two decades, the internet was a research network — connecting universities, government labs, and defense facilities.

It was expensive to access, required technical expertise to use, and served a community of researchers who governed it through informal consensus.

The technology was concentrated not because anyone restricted it but because nobody else had a reason to use it.

Phase 2 — Democratization.

The inflection point was 1993 — the year Mosaic, the first graphical web browser, was released by a team at the University of Illinois.

For the first time, a person without technical training could navigate the World Wide Web by clicking on links.

The web had existed since Tim Berners-Lee's 1989 proposal, but Mosaic made it usable.

Within a year, web traffic had increased by 341,634%.

The browser did for the web what the printing press did for text: it made the technology accessible to anyone.

The second inflection was 2007 — the year Apple released the iPhone.

The smartphone put the internet in every pocket.

By 2015, more than two billion people had smartphones.

By 2020, more than four billion.

The democratization was not gradual.

It was an explosion — the fastest adoption of a technology in human history.

The printing press took fifty years to reach 250 cities.

The smartphone reached four billion people in thirteen years.

> *"The browser did for the web what the printing press did for text: it made the technology accessible to anyone. The smartphone completed the revolution. Four billion people in thirteen years — the fastest adoption of a technology in human history."*

Phase 3 — Chaos.

The chaos did not arrive as a single event.

It accumulated — gradually at first, then all at once, across multiple domains simultaneously.

In commerce, the internet enabled new forms of fraud, identity theft, and consumer exploitation that existing law was not designed to address.

In communication, it enabled harassment, stalking, and abuse at a scale and speed that overwhelmed every moderation system.

In information, it enabled the mass production and distribution of disinformation — fake news, conspiracy theories, fabricated images, and manufactured outrage — at a cost approaching zero.

In politics, it enabled foreign interference in elections, the micro-targeting of voters with psychologically manipulative advertising, and the algorithmic amplification of political extremism.

In privacy, it enabled the collection of personal data at a scale no previous surveillance system had achieved, by companies whose business model depended on knowing everything about their users and selling that knowledge to advertisers.

And in the most extreme case — documented in the first book of this trilogy — it enabled a social media platform's recommendation algorithm to amplify content that a United Nations fact-finding mission determined played a "determining role" in inciting genocide against the Rohingya people of Myanmar.

The system was not broken.

The system was optimizing for engagement. Engagement and incitement produced the same signal.

Phase 4 — Consolidation. While the chaos was unfolding, the consolidation was proceeding with remarkable efficiency.

By 2020, five companies — Google, Amazon, Facebook, Apple, and Microsoft — controlled the majority of the internet's core functions.

Google controlled search — the gateway through which most people accessed information.

Amazon controlled e-commerce and, through Amazon Web Services, the cloud infrastructure on which a significant portion of the internet itself runs.

Facebook controlled the social graph — the map of human relationships and the primary channel through which billions of people received news, shared experiences, and formed opinions.

Apple controlled the device — the smartphone through which most people accessed everything else, and the App Store that determined which software could reach those devices.

Microsoft controlled the enterprise layer — the office software, cloud services, and operating systems on which businesses ran.

The distribute-then-recapture pattern traced across five thousand years of computation in *Who Holds the Jar?*, operating in real time.

The internet distributed capability — anyone could publish, sell, communicate, organize. Five companies recaptured the infrastructure, the distribution, and the data.

The users had access.

The companies had control.

And the difference between access and control is the difference between using a road and owning the road.

> *"The users had access. The companies had control. The difference between access and control is the difference between using a road and owning the road."*

Phase 5 — Governance.

The governance arrived late, fragmented, and outpaced by the technology it sought to govern.

The European Union's General Data Protection Regulation — GDPR — came into effect on May 25, 2018.

It was the first comprehensive regulatory framework for digital privacy and data protection.

It arrived thirty years after the World Wide Web was proposed, forty-nine years after ARPANET, and twenty-five years after the Mosaic browser made the web accessible.

By the time GDPR arrived, the consolidation was complete.

The five companies were entrenched.

The surveillance architecture was built.

The recommendation algorithms were running.

The data had been collected, processed, sold, and weaponized for a quarter of a century before the first major regulatory framework attempted to constrain any of it.

The United States, the nation that built the internet and hosts the companies that consolidated it, has not, as of this writing, passed a comprehensive federal privacy law. Section 230 of the Communications Decency Act — passed in 1996, when fewer than 40 million people used the internet — remains the primary legal framework governing platform liability.

It was written for a world of small bulletin boards and personal web pages.

It now governs platforms that serve billions of users and whose algorithmic decisions shape elections, public health, and the information environment of entire nations.

The Lesson

The internet's governance failure is not a story about bad actors.

It is a story about the absence of a room.

No equivalent of the Constitutional Convention gathered to write the rules for how the internet would be governed.

No equivalent of Bretton Woods convened to determine how the economic power the internet generated would be distributed.

No equivalent of the Geneva Conventions established rules for how the internet's capacity for harm would be constrained.

No equivalent of the NPT was signed to prevent the concentration of internet power in a small number of entities.

In the absence of a room, the rules were written by the consolidators.

The terms of service that govern the speech, commerce, and associations of billions of people were drafted by corporate lawyers in Mountain View and Menlo Park.

The algorithms that determine what billions of people see, read, and believe were designed by engineers optimizing for engagement.

The data collection practices that have mapped the intimate details of billions of lives were implemented by companies whose revenue depends on selling access to that map. These are governance decisions.

They determine who can speak and who is silenced.

They determine which information is amplified and which is suppressed.

They determine whose data is collected, how it is used, and who profits from it.

They are governance decisions made without democratic mandate, without public deliberation, without accountability to the people who live under them, and without the legitimacy that comes from a process in which the governed have a voice.

The internet's lesson is not that technology is ungovernable.

The printing press was governed. Electricity was governed. Nuclear energy was governed.

Each technology was more powerful and more rapidly deployed than the last, and each was brought under a governance framework that, however imperfect, balanced the technology's benefits against its risks.

The internet was not governed because the room did not convene — not because governance was impossible, but because the prevailing ideology of the 1990s and 2000s held that the internet was best governed by no one.

> *"The internet was not governed because the room did not convene. Not because governance was impossible, but because the prevailing ideology*

held that the internet was best governed by no one. The ideology was tested. It failed.”

The Ideology

The ideology that prevented the room from convening has a name.

It was called internet exceptionalism — the belief that the internet was fundamentally different from every previous technology and therefore required a fundamentally different approach to governance, which in practice meant no governance at all.

The argument took several forms.

The internet was global and therefore could not be governed by national regulators.

The internet was decentralized and therefore could not be controlled by any single authority.

The internet was a force for freedom and therefore regulation would inevitably be censorship.

The internet moved too fast for regulation to keep up, and therefore regulation would inevitably stifle innovation.

The market would self-correct — bad platforms would lose users, good platforms would thrive, and competition would produce better outcomes than any regulator could mandate.

Each of these arguments contained a kernel of truth.

The internet is global.

It is (or was) decentralized.

It did enable remarkable freedoms. Badly designed regulation can stifle innovation. Competition can improve outcomes.

But the arguments were used not as cautions to guide governance but as justifications to prevent it.

The kernel of truth in each argument was stretched into an absolute: because governance is difficult, governance should not be attempted.

Because regulation might be imperfect, regulation should not exist.

Because the internet is different from previous technologies, the lessons of previous technologies do not apply.

The lessons applied.

The five-phase cycle played out exactly as it had for every prior technology. Innovation, democratization, chaos, consolidation.

The only variable was Phase 5.

And Phase 5 did not arrive on time — because the people who would have needed to convene the room were persuaded that the room was unnecessary.

The Cost of Thirty Years

The cost of the internet's ungoverned chaos phase is not abstract.

It is documented, quantified, and, in some cases, counted in bodies.

In Myanmar, an ungoverned recommendation algorithm amplified content that contributed to the displacement of 700,000 people and the killing of thousands more.

In the United States, ungoverned data collection practices enabled the Cambridge Analytica scandal — the harvesting of personal data from 87 million Facebook users without their meaningful consent, used to micro-target political advertising in the 2016 presidential election.

In Ethiopia, in India, in Sri Lanka, in the Philippines, social media platforms that operated without adequate content governance amplified communal violence, political extremism, and disinformation campaigns that destabilized democratic institutions.

In the domain of privacy, the ungoverned internet produced what the scholar Shoshana Zuboff named *surveillance capitalism* — a new economic logic in which the prediction and modification of human behavior is the product, personal experience is the raw material, and the extraction of behavioral data at scale is the means of production.

The data that users provided — their searches, their locations, their purchases, their relationships, their conversations, their medical concerns, their political views — was harvested, analyzed, packaged, and sold to advertisers, political campaigns, and, as the Snowden disclosures revealed in 2013, intelligence agencies. None of this required breaking the law.

The law had not been written.

In the domain of labor, the internet's platforms created new categories of precarious work — the gig economy, the algorithmic management of warehouse workers, the content moderation sweatshops in which underpaid workers in the Philippines and Kenya reviewed the most violent and traumatizing content the internet produced, absorbing the psychological cost of maintaining the platforms' usability so that the platforms' users would not have to. These workers had no union, no regulatory protection specific to their conditions, and no seat in any room where the rules were being discussed.

> *“Content moderators in the Philippines and Kenya reviewed the most violent content the internet produced, absorbing the psychological cost of maintaining the platforms’ usability. They had no union, no regulatory protection, and no seat in any room where the rules were being discussed.”*

What the Internet Teaches AI

The internet’s five-phase cycle completed in approximately fifty years — from ARPANET in 1969 to the current state of consolidated, inadequately governed platform power.

This is roughly the same duration as the electricity cycle.

It is significantly faster than the printing press.

And it produced a Phase 5 that arrived too late to prevent the consolidation it was supposed to constrain. AI is following the same cycle.

The parallels are not approximate.

They are precise.

The innovation phase began with the deep learning breakthroughs of 2012–2020 — the moment when neural networks began achieving superhuman performance on specific tasks: image recognition, language translation, game playing.

The capability existed.

It was expensive, concentrated in a handful of research labs, and inaccessible to the general population.

The democratization phase began on November 30, 2022, when OpenAI released ChatGPT to the public.

The system reached 100 million users in two months — the fastest adoption of a consumer technology in history, surpassing even the smartphone.

Within a year, open-source models were available for anyone to download and run.

Within two years, AI code generation tools were producing functional software from natural-language descriptions.

The capability was distributed.

The chaos phase is now. AI-generated disinformation is flooding information channels. AI code generation is producing software with vulnerabilities that the builders cannot detect. AI systems are being deployed in healthcare, criminal justice, military operations, and critical infrastructure without adequate testing, auditing, or oversight.

The harms documented in the first book of this trilogy — biased sentencing, discriminatory healthcare, algorithmic amplification of extremism, AI-assisted targeting — are not hypothetical future risks.

They are current realities.

The consolidation phase is already visible. Seven organizations can train frontier models.

Two companies manufacture the required chips. Five cloud providers control the compute infrastructure.

One company controls over sixty percent of active satellites in low Earth orbit.

The concentration of AI capability and infrastructure is proceeding faster than the consolidation of the internet — because AI's infrastructure requirements are more capital-intensive, more technically demanding, and more geographically concentrated than the internet's ever were.

And the governance phase? Fragments.

The EU AI Act. Executive orders. National strategies. Corporate self-regulation. Voluntary commitments.

No Philadelphia.

No Bretton Woods.

No NPT.

No comprehensive, enforceable, internationally coordinated framework for the governance of artificial intelligence.

> *"The chaos phase is now. The consolidation is already visible. The governance is fragmentary. The internet's lesson is clear: if the room does not convene during the window, the consolidators write the rules. The window for AI is open. It is closing."*

The Difference This Time

There is one crucial difference between the internet's cycle and AI's, and it cuts in both directions.

The internet's chaos phase was, in a sense, novel. Nobody had seen a global communications platform before. Nobody had experienced algorithmic amplification at scale.

The ideology of internet exceptionalism could persist because there was no precedent to contradict it — no previous technology whose ungoverned chaos had produced the specific categories of harm that the internet produced. AI does not have this excuse.

We have the precedent.

The internet *is* the precedent.

Every failure of internet governance — the platform monopolies, the surveillance architecture, the algorithmic amplification of extremism, the erosion of privacy, the displacement of labor without safety nets, the concentration of power in five companies — is a warning, written in the lived experience of four billion people, about what happens when a transformative technology reaches consolidation without governance.

The difference this time is that the excuse is gone.

We cannot claim that no one could have foreseen the consequences.

We have seen them.

We are living in them.

The internet's ungoverned chaos is the air we breathe.

If AI follows the same path — and every indicator suggests it is following the same path, faster — then the failure is not ignorance.

It is choice.

The printing press took 350 years to govern because nobody had seen a printing press before. Electricity took 53 years because nobody had electrified a continent before. Nuclear energy took 23 years because nobody had split an atom before.

The internet was never adequately governed, in part because nobody had built a global communications platform before.

We have built one now.

We know what happened.

The question is whether we will let it happen again.

Part I of this book is complete.

The pattern has been established: five phases, observable across centuries, accelerating with each era, and producing consequences that are determined by the duration of the chaos phase — the period between democratization and governance.

The internet is the warning case: the technology that was never adequately governed, whose chaos we are still living through, whose consolidation has locked in power structures that governance now struggles to dislodge.

Part II enters the chaos phase of artificial intelligence. Not as an abstraction. As a set of stories about specific people, in specific places, experiencing the consequences of a technology that has been democratized without governance.

The new builders who create software they cannot secure.

The information environment drowning in synthetic content.
The workers whose judgment has been declared economically unnecessary.
software this way in the next five years.

* * *

CHAPTER GC-4

The New Builders

On the developer who built a working application in a weekend, the vulnerability they could not see, and the scale of what is coming

The New Builders

Imagine a person — call her Maya — who runs a small physical therapy practice with three employees.

She needs a system to manage patient intake, schedule appointments, and track treatment notes.

The commercial software options cost between $200 and $500 per month, and none of them do exactly what she needs.

She has heard about AI tools that can build applications from plain-language descriptions.

On a Saturday morning, she opens one of these tools and types: “Build me a patient management system for a small physical therapy practice.” By Sunday evening, she has a working application.

It has a patient intake form, a scheduling calendar, a treatment notes interface, and a dashboard.

It looks professional.

It functions. Her employees can use it on Monday.

She is delighted.

She has saved thousands of dollars in software costs and built something tailored to her practice’s specific needs.

The application stores patient health records — protected health information under federal privacy laws — HIPAA in the United States, PIPEDA in Canada, GDPR in Europe — regulations that impose strict requirements on how such data is stored, accessed, and secured.

The AI tool that generated the code did not implement encryption at rest for the database.

It did not implement proper session management.

It did not implement role-based access controls.

It did not include audit logging.

It stored the database connection credentials in a configuration file accessible from the web. Maya does not know what any of these things are.

The AI tool did not tell her they were missing, because the AI tool was not designed to audit its own output for security or regulatory compliance.

It was designed to produce functional code from natural-language descriptions. Functional and secure are not the same thing. Maya is not a software engineer.

She is not supposed to be.

She is a physical therapist who needed a scheduling system.

The tool gave her one.

The tool did not give her the knowledge to evaluate whether what it gave her was safe.

And no governance framework requires her to have that knowledge, requires the tool to provide that evaluation, or requires anyone to stand between the tool's output and the patient data it will process.

In 2025, security researchers identified a critical vulnerability — CVE-2025-48757 — in applications built using Lovable, one of the most popular AI application builders.

The vulnerability exposed the data of applications built on the platform to unauthorized access.

The security assessment found that approximately one in ten applications built with the tool were affected.

One in ten.

The vulnerability was not in the users' code.

It was in the patterns the AI generated — the default configurations, the architectural choices, the security assumptions embedded in the code the tool produced.

The users who built applications with Lovable did not introduce the vulnerability.

They inherited it.

They inherited it from a tool they trusted, in code they could not read, implementing security practices they could not evaluate.

The reader who has followed this trilogy will recognize the architecture.

In *The Last Keeper*, Chapter 3, the Therac-25's software was reused from the Therac-20.

The software contained a flaw that was harmless on the old machine, where hardware interlocks caught it, and lethal on the new machine, where the interlocks had been removed.

In *The Last Keeper*, Chapter 6, the Ariane 5 rocket reused software from the Ariane 4.

The code worked exactly as written — for the wrong rocket.

The architecture recurs: code produced in one context, deployed in another, carrying assumptions that are invisible to the person who deploys it.

The difference in scale is the difference that demands governance.

The Therac-25 was deployed in a handful of clinics.

The Ariane 5 was a single rocket. Lovable and tools like it are used by tens of thousands of builders, producing applications that serve millions of users, handling data that includes medical records, financial information, personal identification, and communications.

The inherited flaw scales with the tool's adoption.

The Therac-25's flaw harmed six patients. A flaw in an AI code generation pattern can potentially affect every application built with that pattern.

The Paradox

In 2025, the nonprofit research organization METR — Model Evaluation and Threat Research — published a study that measured the actual productivity impact of AI coding tools on experienced software developers working on real-world projects.

The finding was striking: developers using AI assistance were 19% slower than developers working without it.

At the same time, the developers using AI assistance reported feeling 20% faster.

The gap between perceived and actual performance is not a curiosity.

It is a safety problem. A developer who believes they are working faster and more effectively will rely more heavily on the tool.

They will review the tool's output less carefully.

They will trust the generated code more readily.

And the code they trust will contain the assumptions, the defaults, and the blind spots of the model that generated it — assumptions and blind spots that the developer, feeling confident and productive, is less likely to question.

> *"Maya is not a software engineer. She is not supposed to be. The tool gave her a working application. It did not give her the knowledge to evaluate whether what it gave her was safe. Nobody stands between the tool's output and the patient data."*

The Vulnerability

Maya is fictional.

The vulnerability is not. *The Last Keeper*, Chapter 4 — the pilots of Air France 447 — translated from the cockpit to the code editor.

The more reliable the autopilot, the less the pilot flew manually, the worse the pilot performed when the autopilot disconnected.

The more capable the AI coding tool, the less the developer exercises independent judgment about the code, the worse the developer performs at catching the tool's errors.

The tool is good enough that the developer stops checking.

The developer stops checking precisely because the tool is good enough.

And then the tool produces a vulnerability, and nobody is watching.

> *"Developers using AI assistance were 19% slower but felt 20% faster. The gap between perceived and actual performance is not a curiosity. It is the automation paradox, translated from the cockpit to the code editor."*

The Mandate

In early 2025, Tobi Lütke, the CEO of Shopify, sent an internal memo to his company that was subsequently made public.

The memo stated that the use of AI tools was now a baseline expectation for every employee at the company. Not an option. Not an experiment. A baseline. Employees would be evaluated on their ability to leverage AI effectively. Teams requesting additional headcount would first need to demonstrate that they had explored what AI could do in place of a new hire. Shopify is one of the largest e-commerce platforms in the world, serving over 2 million merchants globally.

The memo was not a suggestion from a startup founder.

It was a policy directive from the CEO of a company whose infrastructure underpins a significant fraction of global online commerce.

And it was representative of a broader shift: across the technology industry, AI adoption moved in 2025 from exploration to mandate. Companies did not ask employees whether they wanted to use AI.

They told them to.

The implications extend beyond Shopify's employees.

The merchants on Shopify's platform — the small businesses, the independent sellers, the entrepreneurs — would increasingly interact with AI-generated tools, AI-generated interfaces, and AI-generated business logic built by a workforce that had been instructed to rely on AI as a default.

The applications that Maya the physical therapist builds with AI tools are one layer of the problem.

The platforms on which those applications run, themselves built with increasing AI assistance, are another.

The assumption of human review at every layer of the software stack is no longer valid.

The Lego and the Lumber

There is a way to understand the landscape that this chapter describes, and it is the distinction between two approaches to building software that will define the governance challenge for the next decade.

* * *

The automation paradox from the dynamic that degraded

The first approach is model-driven development — building software on platforms that enforce structure, constraints, and guardrails by design.

The bricks snap together in predefined ways. You can build complex structures, but the bricks constrain what you can build to configurations that the platform has validated.

The brick cannot be assembled in a way that produces an insecure database connection, because the platform does not offer that assembly.

The safety is in the brick, not in the builder.

The second approach is code-generated development — building software by generating raw code from AI tools, with no platform enforcing structure or constraints. You can build anything.

The lumber does not constrain you.

But the lumber does not protect you, either.

If you do not know how to frame a load-bearing wall, the house falls down.

If you do not know how to implement encryption at rest, the patient data is exposed.

The capability is in the tool.

The safety is in the builder's knowledge.

And the builder, increasingly, does not have that knowledge — because the tool has made the knowledge feel unnecessary.

> *"Model-driven development is Lego: the safety is in the brick. Code-generated development is raw lumber: the safety is in the builder's knowledge. The tool has made the knowledge feel unnecessary."*

The governance challenge is this: AI code generation is expanding the population of builders at an rate. People who could not build software a year ago can now produce functional applications in hours.

This is genuinely democratizing.

It is genuinely valuable.

And it is genuinely dangerous, because the new builders are building with lumber — raw, unstructured, unconstrained code — without the expertise to know whether what they have built is sound.

The printing press democratized the production of text.

The chaos phase was fueled by the fact that anyone could now publish anything — including inflammatory pamphlets that incited religious warfare. AI code generation is democratizing the production of software.

The chaos phase will be fueled by the fact that anyone can now build anything — including applications that handle sensitive data without the security practices that a professional developer would consider non-negotiable.

The Scale of What Is Coming

Maya's physical therapy practice is one case.

The scale of the phenomenon is vastly larger.

In the United States alone, there are approximately 33 million small businesses. Globally, the number exceeds 400 million.

The vast majority of these businesses do not employ software developers.

They have historically relied on commercial software — purchased applications built by professional development teams, tested for security and compliance, and maintained through regular updates.

The commercial software was expensive and generic.

But it was, in most cases, professionally built. Like building with Lego bricks. Like building with raw lumber. AI code generation changes this equation. Within the next five years, an estimated twenty million or more small businesses, nonprofits, and individuals will use AI tools to build custom software for their specific needs, at a remarkable rate.

They will build patient management systems.

They will build customer databases.

They will build payment processing interfaces.

They will build tools that handle financial records, medical histories, educational data, and personal communications.

They will build these tools because AI has made building possible for people who could not build before.

And they will deploy these tools without security audits, without compliance reviews, without penetration testing, and without the professional oversight that the commercial software they are replacing would have received.

The comparison to the printing press is precise.

Before Gutenberg, the production of text was controlled by trained scribes operating within institutional frameworks — monasteries, universities, royal courts — that provided a degree of quality control and institutional accountability.

After Gutenberg, anyone with access to a press could produce text.

The quality, accuracy, and responsibility of that text became the individual printer's concern. Some printers were careful. Many were not.

The governance frameworks that eventually addressed this — copyright law, libel law, press standards — took centuries to develop.

Before AI code generation, the production of software was controlled by trained developers operating within professional frameworks — engineering teams, code review processes, security audits, compliance certifications — that provided a degree of quality control and institutional accountability.

After AI code generation, anyone with access to the tool can produce software.

The security, compliance, and reliability of that software becomes the individual builder's concern. Some builders will be careful. Many will not know what careful means.

> *“Before AI code generation, software was produced by trained developers within professional frameworks. After AI code generation, anyone can produce software. Some builders will be careful. Many will not know what careful means.”*

The Missing Keeper

In the old model, there were keepers in the software development process. Senior engineers who reviewed code before it was deployed. Security teams that audited applications before they went live. Compliance officers who verified that systems handling regulated data met the applicable requirements. QA testers who looked for the failure modes the developers did not anticipate. These keepers were expensive.

They were slow.

They were, from the perspective of a small business trying to ship a product, an obstacle.

But they were the mechanism by which software moved from “it works” to “it works and it is safe.” The distance between those two statements is the distance between Maya’s application and an application that a professional team would deploy to handle patient health records. AI code generation removes these keepers — not by replacing them with something better, but by making them seem unnecessary.

The tool produces working code. Working code feels like finished code.

The builder who has never worked with a security team does not know that a security review is missing.

The builder who has never seen a compliance audit does not know that a compliance audit is needed.

The absence of the keeper is invisible to the person who has never seen the keeper work.

This is a governance problem, not a technology problem.

The technology is doing what it was designed to do: producing functional software from natural-language descriptions.

The governance question is: who ensures that the software is safe? Who stands between the tool’s output and the data it will process? Who catches the vulnerability that the builder cannot see and the tool does not flag?

In the current landscape, the answer is: nobody.

No regulatory framework requires AI-generated software to be audited before deployment.

No licensing requirement ensures that the person building software with AI tools understands the security implications of what they are building.

No certification standard applies to the AI tool itself, requiring it to produce code that meets minimum security or compliance benchmarks.

No governance framework exists for this scenario.

And twenty million builders are about to build.

> *"No regulatory framework requires AI-generated software to be audited before deployment. No certification standard applies to the AI tool itself. No governance framework exists for this scenario. And twenty million builders are about to build."*

What Governance Would Look Like

This chapter is not an argument against AI code generation.

The democratization of building is genuine and valuable. Maya should be able to build a patient management system that serves her practice.

The question is not whether she should build.

The question is what the governance framework should require so that what she builds does not expose her patients to harm.

The analogy is to building codes — the governance framework that emerged from the chaos phase of physical construction. Anyone can design a house. Not anyone can build one that meets code. Building codes do not prevent people from building.

They ensure that what is built meets minimum standards for safety, structural integrity, and habitability.

They apply to the builder, not to the homeowner's intentions. A homeowner who builds a deck that collapses is responsible for building without a permit. A jurisdiction that does not require a permit is responsible for the absence of the requirement.

The software equivalent of building codes does not exist for AI-generated applications.

It could.

It would require AI code generation tools to include automated security auditing as a default — not as an optional add-on, but as an integral part of the generation process.

It would require applications that handle regulated data to pass automated compliance checks before deployment.

It would require the platforms that host these applications to verify minimum security standards, just as building departments verify minimum construction standards.

It would require the AI tool providers to be accountable for the security patterns their tools produce, just as lumber mills are accountable for the grade of the lumber they sell. None of this is technically impossible. Automated security scanning tools exist. Compliance verification systems exist.

The technology to make AI code generation safer is available. What is absent is the governance framework that requires it to be used.

The building codes have not been written.

The room has not convened.

And every day that the room does not convene, more applications are built, more data is exposed, and more patients, customers, and users bear the risk of a regulatory void they did not create and cannot see.

The new builders are building software without governance.

The next chapter describes a parallel chaos: the new content creators are flooding the information environment with synthetic material that is indistinguishable from reality. AI has democratized not just the production of code but the production of deception — and the cost of determining what is true is rising faster than any institution can bear.

CHAPTER GC-5

The Information Flood

On the democratization of deception, and the epistemic crisis that arrives when fabrication is free and detection is expensive

The Information Flood

In 1835, the *New York Sun* published a series of articles claiming that Sir John Herschel, one of the most prominent astronomers of the era, had observed life on the Moon through a powerful new telescope.

The articles described bat-winged humanoids, blue unicorns, and a civilization of temples and obelisks.

The series, which became known as the Great Moon Hoax, was entirely fabricated.

It was written by a reporter named Richard Adams Locke, and its purpose was to sell newspapers.

It worked.

The *Sun* became the bestselling newspaper in the world.

The Great Moon Hoax required a writer, an editor, a printing press, and a distribution network.

It required weeks of planning and composition.

It required the reputation of a real astronomer to lend credibility.

And it was debunked within weeks by rival publications that had the time, the expertise, and the institutional motivation to investigate the claims. What happens when the cost of producing a convincing fabrication drops to zero and the infrastructure required to debunk it does not change.

The Great Moon Hoax required a newsroom. A deepfake video requires a laptop and a text prompt.

The asymmetry between the cost of producing deception and the cost of identifying it is the structural dynamic that defines the information environment of the AI era — and it is a dynamic that no existing governance framework addresses.

> *"The Great Moon Hoax required a newsroom. A deepfake video requires a laptop and a text prompt. The asymmetry between the cost of producing deception and the cost of identifying it is the crisis."*

The Economics of Deception

Every form of deception has an economics — a cost of production and a cost of detection.

The history of disinformation is the history of that ratio shifting.

Before the printing press, producing a convincing forgery required a skilled scribe, expensive materials, and access to the formats and seals of whatever authority the forgery claimed to represent.

The cost of production was high.

The cost of detection was also high, but the two were roughly in balance — a forgery that took weeks to produce could be investigated over a similar timescale.

The printing press shifted the ratio. Mass-produced pamphlets could spread fabricated claims faster than any authority could respond.

The inflammatory pamphlets that fueled the Wars of Religion were, in their era, the equivalent of viral misinformation — produced cheaply, distributed widely, and consumed by populations that had no institutional mechanism for evaluating their accuracy.

The chaos phase of the printing press was, in significant part, a crisis of information governance. Photography and film shifted the ratio again. A photograph was assumed to be a reliable representation of reality. Manipulating a photograph required darkroom skill and was detectable by experts. A film required actors, sets, cameras, editing equipment, and distribution infrastructure.

The cost of producing convincing visual deception remained high enough that detection could keep pace. Digital editing tools shifted the ratio further. Photoshop made image manipulation accessible to anyone with a computer. Video editing tools made it possible to alter footage.

But professional forensic analysis could still detect the alterations — inconsistencies in lighting, compression artifacts, metadata that revealed the editing history.

The cost of production dropped.

The cost of detection remained manageable. Generative AI has broken the ratio.

The Break

In 2024 and 2025, AI systems capable of generating photorealistic images, realistic video, and natural-sounding audio became widely available — many of them free, open-source, and runnable on a consumer laptop.

The cost of producing a convincing fabrication — a deepfake video of a political figure saying something they never said, a synthetic audio recording of a CEO announcing a merger that does not exist, a fabricated photograph of an event that never happened — dropped from thousands of dollars and specialized expertise to zero dollars and a text prompt.

The cost of detecting these fabrications did not drop.

It increased. Deepfake detection requires specialized AI models trained specifically to identify synthetic content, forensic tools that analyze pixel-level artifacts, and human expertise to interpret the results.

The detection tools lag behind the generation tools — the generators improve faster than the detectors can adapt, because generating convincing content requires solving one problem (make it look real) while detecting it requires solving many (identify every possible artifact of every possible generation method).

The asymmetry is now structural.

One person with a laptop can produce a hundred pieces of synthetic content in a day. A newsroom, a fact-checking organization, or a platform's trust-and-safety team must spend hours or days verifying each one.

The production scales effortlessly.

The detection does not.

And every piece of synthetic content that enters the information environment — whether or not it is detected — raises the ambient level of uncertainty for everything else.

When any image might be fabricated, trust in all images erodes.

When any audio might be synthetic, trust in all audio erodes.

The damage is not limited to the specific fabrication.

The damage is to the epistemic infrastructure itself — the shared capacity of a society to agree on what is real.

The Elections

The 2024 US presidential election cycle was the first major democratic exercise conducted in an information environment saturated with generative AI. Deepfake audio clips of candidates circulated on social media. AI-generated images depicting events that never happened were shared as evidence. Synthetic robocalls mimicking the voice of a sitting president were used to discourage voters from going to the polls in a primary election.

The robocall incident illustrates the asymmetry at human scale.

In January 2024, two days before the New Hampshire primary, between 5,000 and 25,000 registered Democrats received a phone call in which a voice indistinguishable from President Biden's told them not to vote. "Save your vote for November," the synthetic voice said. "Voting this Tuesday only plays into the hands of the Republicans." The call was traced to a political operative who had paid a mere $150 to generate the deepfake audio using a commercially available AI voice-cloning tool.

One hundred and fifty dollars to suppress the votes of thousands of citizens in a presidential primary.

The cost of producing the deception: trivial.

The cost of identifying, investigating, and prosecuting it: months of work by the state attorney general's office, the FCC, and federal investigators.

The United States was not unique.

In elections across the world — in India, Indonesia, the United Kingdom, South Korea, and dozens of other democracies that held elections in 2024 and 2025 — AI-generated content was documented as a tool of political manipulation.

In some cases, the content was produced by domestic political actors.

In others, by foreign governments seeking to influence outcomes.

In many cases, the origin could not be determined — because the tools that produce the content do not leave fingerprints the way a newsroom or a government propaganda bureau does.

The governance response was fragmented and reactive. Some jurisdictions introduced requirements for AI-generated content to be labeled.

The requirements were largely unenforceable — because labeling is voluntary for content shared on open platforms, because the tools that generate the content do not embed mandatory watermarks, and because the platforms that distribute the content have neither the technical capacity nor the economic incentive to verify the provenance of every piece of media that passes through their systems.

This is the content moderation failure, amplified by an order of magnitude.

In *The Last Keeper*, Chapter 8, Facebook's fewer than two dozen Burmese-speaking moderators were overwhelmed by a platform serving twenty million users.

The AI content generation tools now available to anyone can produce synthetic content at a rate that would overwhelm every fact-checking organization on earth working simultaneously.

The Speed-Judgment Tradeoff from that chapter — the system operating faster than any human review can match — has crossed into a domain where the content itself is machine-generated.

The machine produces the content.

The machine distributes the content.

And the human who is supposed to evaluate the content is outpaced at both ends.

> *“When any image might be fabricated, trust in all images erodes. The damage is not limited to the specific fabrication. The damage is to the shared capacity of a society to agree on what is real.”*

The Epistemic Crisis

The term “epistemic crisis” describes a situation in which the shared mechanisms by which a society determines what is true have degraded to the point that consensus on basic facts becomes impossible. Philosophers of science and political theorists have discussed epistemic fragility for decades. What is new is the mechanism by which the crisis is arriving. Previous epistemic crises were produced by the intentional actions of identifiable actors: state propaganda, institutional deception, organized disinformation campaigns.

The actors could, at least in principle, be identified, confronted, and held accountable.

The crisis could be attributed.

The counter could be organized.

The AI-driven epistemic crisis is different in kind.

It is produced not by a coordinated campaign but by the structural economics of information production.

When the cost of producing content approaches zero, the information environment is flooded regardless of anyone’s intent. A teenager generating deepfakes for social media followers. A content farm producing thousands of AI-written articles to capture advertising revenue. A political operative generating a few hundred synthetic images to shift a narrative. An AI chatbot confidently providing medical information that is factually wrong. None of these actors is running a coordinated disinformation campaign. All of them, collectively, are degrading the information environment in ways that no single actor can be held responsible for.

The result is not that people believe lies.

The result is worse: people stop believing anything.

When the information environment is sufficiently polluted, the rational response is not to trust more carefully but to trust less universally. Citizens disengage from news. Voters distrust electoral outcomes. Patients question medical guidance. Juries doubt video evidence.

The foundation of shared factual reality on which democratic governance, public health, and the administration of justice depend begins to dissolve — not because the facts have changed but because the mechanism for agreeing on the facts has been overwhelmed.

> *"The result is not that people believe lies. The result is worse: people stop believing anything. The mechanism for agreeing on facts has been overwhelmed. And when a society cannot agree on what is real, it cannot govern itself."*

The Liar's Dividend

There is a second-order effect that may be more damaging than the fabrications themselves. Researchers have named it the liar's dividend: the benefit that accrues to liars when the existence of deepfakes allows them to dismiss authentic evidence as fabricated. Not a new concept Before deepfakes, a video of a public figure committing an act or making a statement was difficult to deny.

The video existed.

It could be authenticated.

It was evidence.

In the deepfake era, anyone confronted with authentic video evidence can claim it was AI-generated.

The claim does not need to be proven.

It only needs to be plausible — and in an environment where deepfakes are known to exist, it is always plausible.

The liar's dividend inverts the burden of proof.

In a pre-deepfake information environment, the burden was on the person claiming a video was fake to demonstrate the fabrication.

In a post-deepfake environment, the burden shifts to the person presenting the video to prove it is authentic — a proof that may be technically possible for forensic experts but is practically impossible for the average citizen, the journalist on deadline, or the jury evaluating evidence.

The liar's dividend benefits the powerful at the expense of the accountable. A politician caught on video making a damaging statement can claim the video is a deepfake. A corporation whose internal practices are documented on video can challenge the footage's authenticity. A government whose security forces are filmed committing atrocities can dismiss the evidence as AI-generated propaganda.

In each case, the existence of deepfake technology provides plausible deniability that did not previously exist — not because the specific evidence is fabricated, but because the environment in which it is presented no longer assumes that video evidence is real.

The keeper — the journalist, the fact-checker, the forensic analyst, the investigative body — can still do the work.

The work still produces accurate results.

But the environment in which the results are presented has been degraded to the point where the results are dismissible.

The keeper is not removed from the loop.

The loop itself has been made unreliable.

> *"The liar's dividend inverts the burden of proof. Anyone confronted with authentic evidence can claim it was AI-generated. The claim does not need to be proven. It only needs to be plausible. In the deepfake era, it is always plausible."*

The Governance Gap

The governance challenge here is not content moderation — the approach the internet's platforms have pursued, with limited success, for two decades. Content moderation addresses the symptom: harmful content that has already been produced and distributed.

The governance challenge of generative AI addresses the infrastructure: the tools that produce the content, the models that generate it, and the absence of any framework that governs their use. Several governance approaches have been proposed. Content provenance systems — technologies that embed verifiable metadata in authentic media at the point of capture, allowing downstream consumers to verify that an image or video was captured by a real camera and has not been altered — are The keeper's problem restated technically promising.

The Coalition for Content Provenance and Authenticity — known as C2PA — has developed an open standard for this purpose.

But adoption requires cooperation across the entire media chain: camera manufacturers, software developers, social media platforms, and news organizations.

No governance framework mandates adoption.

The standard is voluntary.

And voluntary standards in an environment where the economic incentive favors speed over verification have a poor track record. Mandatory watermarking of AI-generated content has been proposed and, in some jurisdictions, enacted.

The technical challenge is that watermarks can be stripped or degraded by reprocessing the content — screenshotting an image, re-encoding a video, or passing text through a paraphrasing tool. Watermarking addresses content produced by compliant tools.

It does not address content produced by tools that are open-source, self-hosted, or operated by actors who choose not to comply.

The governance reaches the willing.

The willing are not the problem.

The deeper governance challenge is the one this book has traced since the printing press: the technology has democratized a capability, the capability produces consequences that existing governance cannot contain, and the consolidation of the tools in a small number of companies creates a structural tension between governing the tool (which the companies resist) and governing the content (which is impossible at scale).

No governance framework addresses AI-generated information any more than it convened for the internet.

And the chaos is accumulating at a pace that makes the internet's thirty-year governance gap look leisurely.

The Parallel

The previous chapter described the chaos in software: millions of people building applications without the knowledge to evaluate their safety.

This chapter describes the chaos in information: billions of people navigating an environment in which the tools to create convincing content have been given to everyone and the tools to verify that content have been given to almost no one.

The two are not separate problems.

They are the same problem — the Phase 3 chaos of AI's five-phase cycle, expressing itself simultaneously across multiple domains.

In software, the chaos takes the form of ungoverned code.

In information, the chaos takes the form of ungoverned content.

In both cases, the mechanism is identical: a capability that was previously controlled by professionals within institutional frameworks has been democratized without the governance frameworks that made the professional version safe.

The next chapter completes the triptych. Software. Information.

And now: labor.

The third domain in which AI's democratization is producing chaos — not because the technology is failing, but because the technology is succeeding, and the success is displacing the people whose judgment the previous books argued was irreplaceable.

CHAPTER GC-6

The Power Shift

On the displacement of judgment, and the governance vacuum for a labor transformation without precedent

The Power Shift

In 2024, Klarna, the Swedish financial technology company, announced that its AI systems were performing work that had previously required approximately 700 full-time employees.

The announcement was made not with regret but with pride — as evidence of the company's technological sophistication and operational efficiency.

The AI handled customer service inquiries, processed claims, resolved disputes, and performed the routine analytical work that had been the daily labor of seven hundred human beings. Seven hundred is a number. Let us pause on what seven hundred means. Seven hundred people woke up each morning and went to work.

They had skills they had developed over years.

They had expertise in Klarna's products, its policies, its customer base.

They understood the nuances that distinguish a legitimate complaint from a fraudulent one, a confused customer from an angry one, a billing error from a billing dispute.

They exercised judgment — the specific human capability that this trilogy has argued, across two books and twenty-six chapters, is irreplaceable in consequential domains.

And they were replaced. Not by better-trained replacements. Not by a reorganization that shifted their work elsewhere.

By a system that their employer determined could do their work at a fraction of the cost, at greater speed, with acceptable quality.

The announcement did not name the seven hundred.

It did not describe their transition — whether they were reassigned, retrained, or simply let go.

The number was a data point in a corporate efficiency narrative.

The people behind the number were invisible.

> *“Seven hundred people exercised judgment every day. They understood the nuances that distinguish a confused customer from an angry one, a billing error from a dispute. They were replaced by a system that their employer determined could do the work faster and cheaper. The people behind the number were invisible.”*

The Difference This Time

Every major technology transition in the history of industrialization has displaced labor.

The mechanization of agriculture eliminated millions of farm jobs.

The factory system displaced artisans and cottage workers.

The assembly line concentrated production and eliminated skilled trades. Containerization destroyed the longshoreman’s craft. Computerization automated clerical work.

In each case, the transition was painful, often brutal, and eventually — over decades — the labor market adapted. New industries absorbed the displaced workers, or their children, or their grandchildren.

The adaptation was not kind.

It was not fast.

But it happened.

The standard reassurance about AI displacement follows this script: technology has always displaced jobs and always created new ones.

The printing press destroyed the scribe’s profession and created the publishing industry.

The automobile destroyed the horse-drawn carriage industry and created the automotive sector. Computerization destroyed typing pools and created the software industry. AI will destroy some jobs and create others.

The transition will be painful but temporary.

This reassurance is based on a pattern that may not hold for this one.

The reason is a difference that the standard reassurance does not address: every previous automation wave displaced hands.

This one displaces judgment.

The mechanization of agriculture displaced physical labor — the act of planting, harvesting, and processing crops by hand.

The assembly line displaced manual skill — the act of shaping, assembling, and finishing products by hand. Computerization displaced clerical work — the act of recording, calculating, and filing information by hand.

In each case, the work that was automated was execution: the performance of a defined task according to defined procedures.

The judgment about what to do, when to do it, and whether the result was correct remained with the human. AI automates the judgment itself. A customer service AI does not merely transcribe the customer's complaint.

It evaluates the complaint, determines the appropriate resolution, and executes it. A legal AI does not merely search for relevant case law.

It analyzes the case, assesses the arguments, and drafts the brief. A medical AI does not merely display the scan.

It reads the scan, identifies the anomaly, and recommends the diagnosis.

The function being automated is not the hand but the mind — not the execution of a decision but the making of it.

> *"Every previous automation wave displaced the hand — the execution of a defined task. AI displaces the mind — the making of the decision. The function being automated is not execution but judgment."*

The Keepers Displaced

The connection to the first book of this trilogy is direct and troubling. *The Last Keeper* spent twelve chapters arguing that human judgment in consequential systems is irreplaceable — that removing the human from the loop produces catastrophe at the margin, that the automation paradox degrades the human's ability to catch the system's errors, and that the keeper must remain.

The argument was about safety.

It was about the patient on the table, the passenger in the aircraft, the defendant in the courtroom.

The labor market is now removing the keepers for a different reason: not safety but economics.

The seven hundred Klarna employees were not removed because the AI was safer than they were.

They were removed because the AI was cheaper.

The economic logic that drives AI labor displacement is indifferent to the safety argument. A company that replaces human judgment with AI judgment is not making a safety decision.

It is making a cost decision.

And the cost calculation does not include the cost documented in *The Last Keeper* — the cost that arrives at the margin, when the system encounters conditions it was not trained for, and the human who would have caught the error is no longer employed.

The safety argument says: keep the human in the loop.

The economic argument says: remove the human from the payroll.

The two arguments are not debating each other in a conference room.

They are operating simultaneously, in the same organizations, on the same systems, with the economic argument winning consistently because the economic argument produces measurable quarterly savings and the safety argument produces unmeasurable future risk.

The savings are certain and immediate.

The risk is probabilistic and deferred.

In every organization governed by quarterly metrics, the certain and immediate wins.

> *"The safety argument says: keep the human in the loop. The economic argument says: remove the human from the payroll. The savings are certain and immediate. The risk is probabilistic and deferred. In every organization governed by quarterly metrics, the certain wins."*

The Speed

The displacement is not only different in kind from previous waves.

It is different in speed.

The mechanization of agriculture in the United States unfolded over roughly a century — from the mid-nineteenth century to the mid-twentieth.

The transition from horse-drawn farming to mechanized farming took three generations. Workers displaced from farms migrated to cities, found factory work, and their children entered the service economy.

The adaptation was measured in decades and generations.

The computerization of clerical work unfolded over roughly forty years — from the introduction of the personal computer in the early 1980s to the present. Typing pools disappeared. Filing clerks were replaced by databases. Bookkeepers were replaced by spreadsheets.

The displaced workers retrained, found adjacent roles, or aged out of the workforce.

The adaptation was measured in decades. AI labor displacement is measured in months. Klarna's announcement covered a transition that occurred within a single year.

The capabilities of AI systems are improving not on annual release cycles but on quarterly or monthly ones. A task that AI cannot perform in January may be automated by June. A role that is secure in Q1 may be eliminated in Q3.

The pace of displacement exceeds the pace of any institutional response — retraining programs, educational pipelines, social safety nets, labor market adjustments — by an order of magnitude.

No governance framework exists for labor displacement at this speed. Previous transitions, painful as they were, unfolded slowly enough that institutions could adapt. Labor unions formed. Governments enacted minimum wage laws, unemployment insurance, job retraining programs. Social safety nets were built — imperfectly, inadequately, but built — over the decades that the transitions required. The convergence that the title names — AI displacement — does not provide decades.

It provides months.

And the governance response, which moves at the speed of legislation, regulation, and institutional adaptation, cannot match the speed of the technology that is producing the displacement.

The Paradox of Democratization

There is a paradox at the center of this chapter, and it connects directly to the argument of *Who Holds the Jar?*. AI is simultaneously democratizing capability and concentrating economic value. Maya, the physical therapist from Chapter 4, can now build software she could not build before. A student can now produce analysis that previously required a team of analysts. A small business can now generate marketing materials, legal documents, financial models, and customer communications that previously required specialized professionals.

The capability has been distributed. Anyone can do what previously required expertise.

But the economic value of the capability flows not to the new users but to the companies that provide the tools. OpenAI, Anthropic, Google, Microsoft, and the other companies that build and host AI systems capture the subscription fees, the API revenue, the enterprise contracts, and the data generated by billions of interactions.

The professionals whose expertise has been democratized — the analysts, the designers, the writers, the rs, the customer service agents — have seen their economic value diminish, because the scarcity that justified their compensation has been eliminated.

The tool that gives everyone the capability takes away the premium that the capability once commanded. It is the distribution-followed-by-reconcentration pattern from *Who Holds the Jar?*, operating at the level of human labor.

The capability distributes.

The economic value recaptures.

The users feel empowered.

The infrastructure owners accumulate the wealth.

And the professionals who previously held the capability as a source of livelihood find that their expertise has been commoditized by the same tool that made it accessible to everyone else.

> *"AI simultaneously democratizes capability and concentrates economic value. The capability distributes. The wealth recaptures. The users feel empowered. The professionals find their expertise commoditized."*

The Professions

Klarna's customer service agents were the most visible displacement.

They will not be the most consequential.

In April 2025, Shopify CEO Tobi Lütke sent an internal memo that leaked within hours.

The message was blunt: before any team could request additional headcount, they would first need to demonstrate that AI could not do the job.

The memo reframed hiring itself as a failure of imagination — the default was now that the work should be done by the machine, and a human was the exception that required justification. Lütke was not describing a future policy.

He was describing the present.

Within weeks, other technology companies adopted similar frameworks, and the phrase “prove the machine can’t” entered the vocabulary of workforce planning across the industry.

In law, AI systems are already capable of reviewing contracts, conducting legal research, drafting briefs, and summarizing depositions — tasks that constitute the daily work of junior associates and paralegals at law firms worldwide.

The profession that trains its young practitioners by assigning them precisely this work is discovering that the training pipeline and the automation target are the same thing.

The junior lawyer who would have spent three years reviewing contracts — and in doing so, learning to think like a lawyer — will not have those three years.

The work will be done by the machine.

And the senior lawyer who was produced by that training pipeline will not be produced by the next one.

In medicine, AI diagnostic systems are approaching or exceeding the accuracy of human specialists in specific domains — radiology, dermatology, pathology.

The radiologist who spent a decade developing the pattern recognition skills that allow them to read a scan and see the anomaly that others miss is being told that the machine can do it faster and, in controlled studies, more accurately.

The studies may be correct.

The question — examined across *The Last Keeper*, — is what happens when the machine is wrong and the radiologist who would have caught the error has not been trained, because the machine made the training seem unnecessary.

In finance, AI systems are conducting analysis, generating reports, making trading decisions, and assessing risk at speeds and volumes that human analysts cannot match.

In journalism, AI is producing news summaries, sports recaps, earnings reports, and weather forecasts.

In education, AI tutoring systems are providing personalized instruction.

In architecture, AI is generating building designs.

In accounting, AI is preparing tax returns and conducting audits.

The pattern is consistent across every profession: the work that is most amenable to AI automation is the work that constitutes the training pipeline for the profession’s future practitioners.

The junior work — the document review, the preliminary analysis, the first-draft writing, the routine diagnostics — is both the most automatable and the most formative. Automating the junior work accelerates the present and hollows out the future.

The firm is more efficient today.

The firm has no pathway to develop the senior expertise it will need in ten years.

> *"The work most amenable to AI automation is the work that constitutes the training pipeline for the profession's future practitioners. Automating the junior work accelerates the present and hollows out the future."*

The Governance Vacuum

When the mechanization of agriculture displaced millions of farm workers, the governance response — slow, partial, and often inadequate — included agricultural subsidies, rural development programs, the expansion of public education to prepare the next generation for industrial work, and eventually the social safety net programs of the New Deal. These responses took decades to develop and never fully addressed the displacement.

But they existed.

They were written.

The room convened, however belatedly.

When computerization displaced clerical workers, the governance response included workforce retraining programs, community college expansion, and the development of technology education in public schools. Again, inadequate. Again, slow.

But present.

For AI-driven labor displacement, as of this writing, the governance response is: almost nothing.

No major economy has enacted legislation specifically addressing AI labor displacement.

No comprehensive retraining framework has been established at a scale proportional to the displacement that is already occurring.

No social safety net adaptation has been designed for a labor market in which the pace of job elimination exceeds the pace of job creation for the first time in the history of industrialization.

The proposals that exist — universal basic income, job guarantee programs, AI taxation to fund transition support — remain proposals.

They are discussed in academic papers, debated at conferences, and advocated by think tanks.

They are not law.
They are not policy.
They are not governance.
The governance response is: almost nothing.
And the seven hundred are already gone.

The Chaos Phase Restated

The chaos phase of artificial intelligence has now been traced across three domains.

In Chapter 4, the chaos in software: millions of people building applications without the expertise to evaluate their safety, using tools that produce functional code without producing secure code, in a landscape where no governance framework requires the tools or the builders to meet minimum standards.

In Chapter 5, the chaos in information: the cost of producing convincing fabrication has dropped to zero while the cost of detecting it has not, producing an epistemic crisis in which the shared capacity of societies to agree on basic facts is being overwhelmed, and the liar's dividend allows the powerful to dismiss authentic evidence as fabricated.

In this chapter, the chaos in labor: AI is displacing not hands but judgment, at a speed that exceeds the capacity of any institutional response, hollowing out the training pipelines that produce the next generation of expertise, and concentrating the economic value of the democratized capability in the companies that provide the tools.
Software. Information. Labor.

Three domains.

One pattern: a capability has been democratized without the governance frameworks that would make the democratization safe for the people it affects.

The tools are available.

The rules are not.

The chaos is the gap between the two.

Part III asks the question that follows: in the absence of governance, who is consolidating control? Who is gaining the power that the chaos phase produces?

And what are they doing with it?

> *"Software. Information. Labor. Three domains. One pattern: a capability has been democratized without the governance frameworks that would make the democratization safe. The tools are available. The rules are not. The chaos is the gap."*

CHAPTER GC-7

The New Gatekeepers

On the consolidation of infrastructure, the concentration of power, and the entities that now control the gates

> *"A single private individual's decisions about a commercial satellite network are now a primary variable in a major conventional war. No governance framework covers this situation. No governance framework was designed to."*

The New Gatekeepers

In September 2022, six Ukrainian drone submarines packed with explosives were navigating through the dark waters off the Crimean Peninsula toward the Russian Black Sea Fleet anchored at Sevastopol.

The drones were guided by Starlink — the satellite internet constellation operated by SpaceX, which is owned by Elon Musk. Ukrainian military planners had designed the operation to strike the fleet that had been bombarding Ukrainian coastal cities and enforcing a naval blockade for seven months.

The drones lost connectivity.

They washed ashore harmlessly.

The Russian fleet survived. Musk had refused to extend Starlink coverage to Crimea.

He later stated that activating Starlink around Sevastopol would have made SpaceX complicit in a major act of war.

He had spoken with the Russian ambassador to the United States, who had told him that a Ukrainian attack on Crimea could provoke a nuclear response.

The decision to deny coverage was made by one person. Not a head of state. Not a military commander. Not an elected official of any government. A private citizen who owned the infrastructure.

The Ukrainian perspective on the decision was captured by a participant in the operation, who recalled the moment communication was cut: the team was seventy kilometers from the target, the operation was underway, and the connectivity disappeared. Ukrainian officials attempted to persuade Musk to restore coverage. American officials were contacted.

The response from the Americans: it was a private company, and they could not put pressure on it. Mykhailo Podolyak, a senior advisor to President Zelenskyy, summarized the consequence: by preventing the Ukrainian drone attack, the decision allowed the Russian fleet to continue firing Kalibr cruise missiles at Ukrainian cities.

The fleet that was not sunk that night continued to bombard civilian infrastructure for months afterward.

> *"The drones lost connectivity. The fleet survived. One person made the decision. Not a head of state. Not a military commander. A private citizen who owned the infrastructure. The fleet he preserved continued bombarding Ukrainian cities for months."*

The Switch

Three and a half years later, in February 2026, the switch flipped in the other direction. Evidence had emerged that Russian forces were using Starlink terminals — obtained through third-party countries — to guide long-range strike drones into Ukrainian cities. A Russian BM-35 drone had penetrated central Kyiv, flying so low that officials in the Cabinet of Ministers building could watch it pass beneath them from the seventh floor. Ukrainian Defense Minister Mykhailo Fedorov presented Musk with evidence of the Russian military's use of his network.

Musk cut off unauthorized Russian access to Starlink.

The effect was immediate and devastating.

Within five days, Ukrainian forces reclaimed over two hundred square kilometers of territory — roughly equivalent to Russia's gains for the entire previous month. Russian frontline communications collapsed. Units could not coordinate.

In one incident on the Zaporizhzhia front, twelve Russian soldiers were killed by friendly fire after a Starlink terminal failure disrupted their coordination. Russia's own satellite communications system, operated by Gazprom Space Systems, was regarded as far less reliable and could not fill the gap. Ukrainian cyber operatives, exploiting the chaos, built a fake Starlink registration portal and invited Russian soldiers to register terminals for a fee.

The scam worked: Russia's 256th Cyber Assault Division collected the precise locations of 2,420 enemy Starlink terminals in a single week — data that was fed directly to artillery and drone targeting units.

Two hundred square kilometers. Twelve soldiers killed by friendly fire. 2,420 terminals exposed.

Because one person flipped a switch on a commercial satellite network. Not a weapon. Not a military system. A commercial internet service that both sides of a war had incorporated into their military operations because no alternative of comparable capability existed.

> *"Two hundred square kilometers changed hands. Twelve soldiers killed by friendly fire. 2,420 terminals exposed. Because one person flipped a switch on a commercial satellite network. The infrastructure of war shifted because one person decided it should."*

The Dependency

The European response to the February 2026 events was not celebration.

It was alarm.

The alarm was not about the decision itself but about the dependency it revealed. European defense officials recognized that the same system that cut Russia off in February could theoretically cut Ukraine off in March.

The switch belonged to one person.

The governance belonged to no one. As one European defense industry executive stated: Europe cannot build a military strategy that runs through one man's decisions. Starlink operates approximately seven thousand satellites — over sixty percent of all active satellites in low Earth orbit.

In Ukraine, over 42,000 Starlink terminals are used by the military, hospitals, businesses, and government agencies. During Russian bombing campaigns that caused widespread blackouts, Ukraine's public agencies turned to Starlink to stay online.

The dependency is not incidental.

It is structural.

No alternative of comparable capability, coverage, and resilience exists. Europe's own satellite communications programs are years from operational deployment at equivalent scale.

The US Air Force Secretary, responding to the original Crimea incident, articulated the problem in terms that apply to every domain this chapter examines: “If we’re going to rely upon commercial architectures or commercial systems for operational use, then we have to have some assurances that they’re going to be available. Otherwise they are a convenience and maybe an economy in peacetime, but they’re not something we can rely upon in wartime.” The assurances do not exist.

No governance framework governs the obligations of a private satellite constellation operator during armed conflict.

No treaty addresses the scenario in which both sides of a war depend on the same commercial infrastructure controlled by a single private individual.

No international body has jurisdiction over orbital communications monopolies.

No framework exists.

And a war is being fought on infrastructure that belongs to a person who is neither a party to the conflict nor accountable to any of the governments whose forces depend on his network.

The honest accounting requires acknowledging the other side.

No government, no international consortium, no distributed democratic process deployed seven thousand satellites in low Earth orbit.

One company did.

The connectivity Ukraine depended on in 2022 existed because of the concentration of capability this chapter warns about. A distributed, democratically governed satellite network would have taken a decade to build and would not have been available when Ukraine needed it.

The tradeoff is real: concentration enables speed; distribution enables governance.

The question is not whether concentration has benefits — it manifestly does.

The question is whether the benefits justify a structure in which one person’s decisions determine the trajectory of a war.

> *“No governance framework governs the obligations of a private satellite operator during armed conflict. No treaty addresses the scenario. No international body has jurisdiction. A war is being fought on infrastructure that belongs to a person who is accountable to none of the governments involved.”*

The Compute Bottleneck

Starlink is the most dramatic example of infrastructure consolidation, but it is not the most consequential for the governance of artificial intelligence.

That distinction belongs to the compute bottleneck. Training a frontier AI model — a system capable of the broad, general-purpose reasoning demonstrated by the most advanced chatbots, code generators, and analytical tools — requires three things: vast quantities of training data, specialized hardware (graphics processing units or tensor processing units, configured in clusters of thousands), and months of continuous computation.

The cost of training a single frontier model is measured in hundreds of millions of dollars.

The number of organizations on Earth with the financial resources, technical expertise, and hardware access to train a frontier model is approximately seven: OpenAI, Anthropic, Google DeepMind, Meta, Mistral, xAI, and a small number of Chinese labs including Baidu and Alibaba's research divisions. Seven organizations.

In a world of eight billion people, 195 nations, and millions of institutions, the capacity to produce the most consequential technology of the twenty-first century is concentrated in fewer than ten organizations, most of them headquartered within a fifty-mile radius of each other in the San Francisco Bay Area.

The concentration deepens at each layer of the stack.

The specialized chips required for AI training are designed primarily by one company: NVIDIA.

They are manufactured primarily by one company: TSMC, based in Taiwan.

The cloud infrastructure on which most AI workloads run is provided by three companies: Amazon Web Services, Microsoft Azure, and Google Cloud.

The pipeline from chip design to chip fabrication to cloud deployment to model training to model serving passes through a remarkably small number of chokepoints — narrow control points in the supply chain where a single entity determines what passes through, each controlled by a single entity or a small oligopoly.

> *"Seven organizations can train frontier models. One company designs the chips. One company manufactures them. Three companies host the cloud. The pipeline from chip to model passes through a remarkably small number of chokepoints, each controlled by a single entity."*

The Pattern Recognized

The architecture is the one traced across five thousand years in *Who Holds the Jar?*.

The mainframe era concentrated computing power in a handful of companies — IBM above all.

The personal computer distributed capability to millions — and then Microsoft and Intel recaptured the value through control of the operating system and the processor.

The internet distributed communication to billions — and then five companies recaptured the infrastructure.

The cycle repeats: distribute the access, recapture the control.

Each era feels like democratization.

Each era produces reconcentration. AI is following the pattern with a new feature: the reconcentration is happening *simultaneously* with the democratization. ChatGPT was released to the public in November 2022.

By the time it reached 100 million users two months later, the companies that built it had already secured the compute, the data, and the distribution channels that would ensure the value flowed to them.

The previous cycles took decades to move from democratization to reconcentration. AI is doing it in real time.

The users experience the distribution.

The companies capture the value.

The two happen concurrently, and the gap between them is closing before any governance framework can pry it open.

The Stack

To understand the full scope of the consolidation, it helps to visualize the AI infrastructure as a vertical stack — a series of layers, each built on the one below it, each controlled by a small number of entities.

At the bottom: the chips. NVIDIA designs the GPUs that train AI models. TSMC manufactures them.

The design and fabrication of the most advanced AI chips are concentrated in two companies, separated by the Pacific Ocean, in a geopolitical relationship (US-Taiwan-China) that is among the most volatile on earth. A disruption to either company — through conflict, natural disaster, or sanctions — would constrain the global AI industry at its foundation.

Above the chips: the compute.

The chips are assembled into data centers operated by cloud providers — primarily AWS, Azure, and Google Cloud. These three companies control the majority of the world’s AI-capable compute infrastructure. A company or government that wants to train an AI model but does not own a data center must rent compute from one of these three providers.

The providers set the prices, the terms of service, and the acceptable use policies.

They are, in effect, the landlords of the AI economy.

Above the compute: the models.

The frontier AI models are built by approximately seven organizations. These models are the engines that power every downstream application — from chatbots to code generators to diagnostic tools to targeting systems.

The organizations that build them determine their capabilities, their limitations, their safety properties, and — through acceptable use policies and content filters — their behavior.

The model builders are not just technology companies.

They are governance actors, whether they acknowledge it or not, because the decisions they make about what their models can and cannot do are decisions that affect every application built on top of them.

Above the models: the distribution.

The models reach users through app stores, API access, enterprise agreements, and integration into existing products (Microsoft’s integration of OpenAI’s models into Office, Google’s integration of Gemini into Search).

The distribution channels are controlled by the same companies that control the compute and, in some cases, the models themselves.

The vertical integration is remarkable: a single company can own the cloud infrastructure, build the model, and distribute it through its own products.

And wrapping around everything: the communications layer.

The orbital infrastructure that connects the user to the service. Sixty percent of active satellites in low Earth orbit belong to one company.

The physical layer on which everything else depends — the ability to reach the cloud, access the model, use the application — passes through a chokepoint controlled by a single individual whose decisions, as this chapter opened by documenting, can determine the outcome of a war.

The Governance Question

The consolidation of AI infrastructure is not a market failure in the traditional sense.

It is a structural consequence of the technology's requirements. Training frontier models requires enormous capital investment, specialized hardware, and concentrated expertise.

The barriers to entry are not artificial — they are the physics and economics of the technology itself.

No antitrust action can make frontier AI training cheap.

But the concentration of capability in a small number of entities creates a governance problem that no existing framework addresses.

The entities that control the AI stack are not regulated utilities.

They are not common carriers.

They are not subject to the obligations that apply to companies whose infrastructure is deemed essential to the public interest.

They set their own terms.

They determine their own acceptable use policies.

They decide, without democratic input, what their models can and cannot do.

When Samuel Insull's utility empire collapsed in the 1930s — as described in Chapter 2 — the governance response was the Public Utility Holding Company Act: regulation that recognized electrical infrastructure as essential and imposed obligations commensurate with that status.

The AI infrastructure is essential.

The cloud compute that runs AI workloads also runs healthcare systems, financial markets, government services, and critical infrastructure.

The orbital communications layer connects military operations, hospitals, and governments in conflict zones.

The models themselves are being embedded in medical diagnosis, legal research, educational instruction, and military targeting.

The infrastructure is essential.

The governance treats it as optional.

The gap between the two is the gap this book exists to name.

> *"The infrastructure is essential. The cloud runs healthcare, finance, and government. The orbital layer connects military operations and hospitals. The models are embedded in diagnosis and targeting. The governance treats all of it as optional."*

The Two Gatekeepers

This chapter has described two forms of gatekeeping, and naming them clearly is essential for the chapters that follow.

The first is infrastructure gatekeeping: control of the physical and computational resources on which everything else depends. TSMC's chip fabrication. NVIDIA's chip design.

The cloud providers' data centers. Starlink's orbital constellation. These gatekeepers control access to the substrate. Without their resources, AI does not function. Without their network, the drone does not reach its target and the hospital does not stay online during a blackout. Infrastructure gatekeeping is control through exclusion: the power to deny access.

The second is capability gatekeeping: control of what the AI can do.

The model builders who determine their systems' safety properties, content policies, and acceptable use restrictions. Anthropic's decision that its models will not be used for autonomous weapons or mass surveillance — traced in *The Last Keeper*, — is an act of capability gatekeeping. So is every content filter, every refusal policy, every safety benchmark. Capability gatekeeping is control through design: the power to shape what the technology does. Both forms of gatekeeping are currently exercised by private companies, at their discretion, without democratic mandate. Musk's Starlink decisions are infrastructure gatekeeping: the power to grant or deny connectivity that determines the outcome of military operations. Anthropic's red lines are capability gatekeeping: the power to determine which uses of AI are permissible. Both decisions may be defensible on their merits. Both decisions were made by private actors exercising judgment that, in a governed world, would be exercised by institutions with democratic accountability.

The question is not whether these gatekeepers are making the right decisions.

The question is whether decisions of this magnitude — decisions that determine the viability of states, the outcome of wars, the boundaries of AI capability — should be made by private actors at all, regardless of how wise or well-intentioned those actors are.

The lighthouse keeper at Smalls was one person watching one light.

The new gatekeepers are private individuals and companies holding switches that affect billions of lives.

The scale demands governance.

The governance does not exist.

> *“The question is not whether these gatekeepers are making the right decisions. The question is whether decisions of this magnitude — decisions that determine the viability of states and the outcome of wars — should be made by private actors at all.”*

The consolidation is domestic and global simultaneously.

The next chapter widens the lens to the geopolitical dimension: the race between nations to control AI capability, the semiconductor export controls that weaponized the compute bottleneck, and the nuclear parallel that provides the only historical precedent for governing a technology of this magnitude at international scale.

CHAPTER GC-8

The Race

On the geopolitical competition for AI supremacy, the weaponization of compute, and the question of whether democracy can compete

> *"The governance is national. The technology is planetary. No nation can govern AI alone. No two nations agree on how to govern it together."*

The Race

On October 7, 2022, the United States Bureau of Industry and Security published a set of export controls that would reshape the global AI landscape.

The controls restricted the sale of advanced semiconductor chips and chip-manufacturing equipment to China.

The restrictions were not targeted at a specific company or a specific product.

They were targeted at a capability: China's ability to train frontier AI models.

The controls leveraged the chokepoints previous chapter. NVIDIA's most advanced AI chips — the A100 and H100 — could no longer be sold to Chinese customers.

The chip-manufacturing equipment produced by ASML, the Dutch company that makes the extreme ultraviolet lithography machines essential for fabricating the most advanced chips, was also restricted through coordinated agreements with the Netherlands and Japan.

The intent was to deny China access not just to the chips themselves but to the machinery needed to build them.

The United States was using its position at the top of the semiconductor stack — the same stack described in Chapter 7 — as a geopolitical weapon.

The compute bottleneck was not merely a market outcome.

It was now a tool of strategic competition.

The country that controlled the chokepoints could determine who had access to the most consequential technology of the twenty-first century and who did not.

> *"The semiconductor export controls were not targeted at a product. They were targeted at a capability: China's ability to train frontier AI models. The compute bottleneck became a geopolitical weapon."*

Two Philosophies

The US-China AI competition is often framed as a technology race — which nation will build the more capable AI system first.

This framing is not wrong, but it is incomplete.

The competition is also, and more fundamentally, a governance race: two superpowers with incompatible visions of how AI should be governed, each building the technology according to its own philosophy, each attempting to ensure that its philosophy prevails.

The Chinese approach integrates AI development with state objectives. China's AI strategy, articulated in the New Generation Artificial Intelligence Development Plan of 2017, envisions AI as a tool of national power: economic competitiveness, military capability, and social management.

The surveillance infrastructure — the Xinjiang monitoring system, the social credit framework, the integrated surveillance platforms — is not a side effect of China's AI described in The Last Keeper described in The Last Keeper development.

It is the development.

The technology and its governance are fused: the state builds the AI, deploys the AI, and determines the AI's purpose.

The governance is total but not democratic.

The rules exist.

They are written by the state, for the state.

The American approach separates AI development from state control — at least in principle.

The United States' AI leadership resides primarily in private companies, funded by private capital, competing in private markets.

The government's role is regulatory rather than directive: setting guardrails (or attempting to) through executive orders, legislation, and agency oversight.

The private companies determine what the AI can do.

The government determines — or fails to determine — what the AI may do.

The separation produces innovation.

It also produces the governance vacuum that seven chapters. Neither philosophy is adequate on its own.

The Chinese model produces governance without liberty — rules written by the state, enforced by the state, serving the state's interests rather than the citizen's.

The American model produces liberty without governance — innovation driven by private actors who operate faster than any regulatory framework can follow, in a political system where the lobbying power of the technology companies exceeds the regulatory capacity of the agencies tasked with governing them.

> *"The Chinese model produces governance without liberty. The American model produces liberty without governance. Neither is adequate. The world needs both."*

The Response

China's response to the semiconductor export controls was not capitulation.

It was acceleration. Chinese AI labs, denied access to the most advanced NVIDIA chips, adapted their training methods to work with less capable hardware.

They developed techniques for training models more efficiently, squeezing more capability from fewer resources. Huawei accelerated the development of its Ascend series of AI chips — less powerful than NVIDIA's best but capable of training competitive models. Chinese companies stockpiled chips purchased before the controls took effect and obtained restricted chips through intermediary countries that were not subject to the export restrictions.

The controls did not stop Chinese AI development.

They may have slowed it.

They certainly changed its trajectory — pushing Chinese labs toward efficiency innovations that American labs, with abundant compute, had less incentive to pursue.

And they accelerated China's drive toward semiconductor self-sufficiency — a program that, if successful, would eliminate the chokepoint the controls were designed to exploit.

The dynamic is familiar to anyone who has studied arms control. Restrictions on a capability create incentives to develop alternative paths to the same capability.

The restriction buys time.

It does not solve the problem.

The problem is that two nations with incompatible governance philosophies are both building documented across the most powerful technology in human history, and no framework exists for governing the competition itself.

The Fragmented Landscape

The United States and China are the dominant competitors, but they are not the only actors.

The global AI governance landscape is a patchwork of national and regional approaches, each reflecting different priorities, different risk assessments, and different political constraints.

The European Union enacted the AI Act in 2024 — the first comprehensive AI regulatory framework in the world.

The Act classifies AI systems by risk level and imposes requirements proportional to the risk: minimal regulation for low-risk applications, strict requirements for high-risk systems used in areas like healthcare, law enforcement, and critical infrastructure, and outright prohibition of certain uses deemed incompatible with fundamental rights, including real-time remote biometric identification in public spaces for law enforcement purposes.

The EU's approach is the most systematic attempt at AI governance to date.

It is also, by the admission of its proponents, a framework designed for the European context — one jurisdiction's rules for a technology that does not respect jurisdictional boundaries.

The United Kingdom has pursued a "pro-innovation" approach — lighter regulation, with sector-specific guidance rather than horizontal legislation.

The philosophy is that heavy regulation would drive AI companies and investment to less regulated jurisdictions. Canada, Japan, South Korea, Singapore, and other nations have published AI strategies, guidelines, and in some cases legislation, each calibrated to domestic political conditions and economic ambitions.

The fragmentation is the problem. AI models are trained in one jurisdiction, hosted in another, accessed from a third, and deployed in all of them simultaneously. A model trained in the United States, hosted on servers in Ireland, accessed by users in Brazil, and deployed in a healthcare system in Nigeria is subject to — in theory — the regulations of every jurisdiction it touches.

In practice, it is subject to the regulations of whichever jurisdiction has the least enforcement capacity, because the companies that build and deploy the models can route around restrictive regulation by hosting in permissive jurisdictions.

The governance is national.

The technology is planetary.

> *"A model trained in the US, hosted in Ireland, accessed from Brazil, deployed in Nigeria is subject to every jurisdiction it touches. In practice, it is subject to the jurisdiction with the least enforcement. The governance is national. The technology is planetary."*

The Rupture

On January 20, 2026, Canadian Prime Minister Mark Carney stood before the World Economic Forum in Davos and delivered a speech that received a rare standing ovation from the assembled heads of state and business leaders.

The speech was, in the assessment of multiple international analysts, the most significant statement by a Western leader on the collapse of the post-Cold War order since the order began to fracture.

Carney's argument was built on a literary reference — Václav Havel's description of how citizens in authoritarian states sustain the system by participating in rituals they privately know to be false.

Carney applied Havel's framework to the rules-based international order: the system's power came not from its truth but from everyone's willingness to perform as if it were true.

Canada and other nations had willingly participated in an order they knew to be partially false — overlooking the unevenness of its application — because they benefited from its predictability. American hegemony had provided genuine public goods: open sea lanes, a stable financial system, collective security, and frameworks for resolving disputes.

That bargain, Carney declared, no longer works. His formulation was precise and unsparing: "We are in the midst of a rupture, not a transition." Great powers had begun using economic integration as weapons — tariffs as leverage, financial infrastructure as coercion, supply chains as vulnerabilities to be exploited.

The integration that had been sold as mutual benefit had become a source of subordination. Nations that negotiated bilaterally with a hegemon negotiated from weakness.

They accepted what was offered.

They competed with each other to be the most accommodating. “This is not sovereignty,” Carney said. “It’s the performance of sovereignty while accepting subordination.”

> *“We are in the midst of a rupture, not a transition. Great powers have begun using economic integration as weapons, supply chains as vulnerabilities, financial infrastructure as coercion.” — Mark Carney, Davos, January 2026*

The passage most relevant to this book — and to the governance challenge it documents — was Carney’s statement on artificial intelligence.

In three sentences, he named the exact convergence that Chapters 7 and 8 of this book have been tracing: “On AI, we are cooperating with like-minded democracies to ensure we will not ultimately be forced to choose between hegemons and hyperscalers.

It is building coalitions that work.” Hegemons and hyperscalers.

Two words that name the two forms of gatekeeping described in Chapter 7: the geopolitical powers that control the strategic environment, and the technology companies that control the infrastructure.

Carney’s formulation acknowledged what the fragmented governance landscape obscures: that the choice facing nations is not between different regulatory approaches but between different forms of dependence. Dependence on a hegemon whose interests may not align with yours. Or dependence on hyperscalers whose infrastructure you cannot replicate and whose terms you cannot negotiate.

The satellite constellation that controls sixty percent of low Earth orbit.

The chip designer.

The chip manufacturer.

The cloud providers.

The model builders.

The stack described in the previous chapter, restated as a geopolitical reality by a sitting head of state. Not naive multilateralism.

Carney’s solution was coalitions — middle powers combining to create what he called “a third path with impact.” Collective investments in resilience, shared standards to reduce fragmentation, buyer’s clubs to diversify supply of critical minerals, and cooperation among like-minded democracies on AI governance.

The approach was neither the American model (liberty without governance) nor the Chinese model (governance without liberty).

It was an attempt to build governance from the middle — from the nations that have the most to lose from a world in which the rules are written exclusively by the powerful. His closing formulation landed with the force of a principle: “Middle powers must act together, because if you’re not at the table, you’re on the menu.” The argument applied not only to nations.

The patients affected by biased healthcare algorithms are not at the table.

The defendants sentenced by proprietary risk scores are not at the table.

The workers displaced by AI systems are not at the table.

The surveilled communities are not at the table.

Carney was speaking about nations, but the principle is universal: absent from the process, present for the consequences.

The missing seats are not only national.

They are individual.

And the cost of absence is borne by whoever is not in the room. Critics have argued that Carney’s simultaneous engagement with Beijing — lowering tariffs on Chinese electric vehicles in exchange for agricultural access — undermines the sovereignty argument by substituting one dependence for another.

The objection is not frivolous.

But the distinction between bilateral negotiation on specific sectors and structural dependence on another power’s infrastructure is precisely the distinction this chapter has been drawing. A trade deal is a policy choice. A satellite constellation you do not own is a dependency you cannot exit.

> *“Middle powers must act together, because if you’re not at the table, you’re on the menu.” — Mark Carney, Davos, January 2026*

The Nuclear Parallel

The closest historical parallel to the current AI governance challenge is the governance of nuclear weapons — the case study from Chapter 2 that demonstrated the fastest progression from innovation to governance in the history of transformative technology.

The parallels are substantial. Nuclear weapons emerged from a national security project during wartime. AI emerged from research labs during a period of intensifying geopolitical competition. Nuclear capability was initially concentrated in one nation and then proliferated to rivals. AI capability is concentrated in a handful of companies in two nations and is proliferating through open-source models, espionage, and parallel development. Nuclear weapons posed an existential risk that no single nation could manage alone. AI poses risks — to labor, to information integrity, to military stability, to surveillance and civil liberties — that no single nation can govern alone.

The NPT, signed in 1968, provided a framework — imperfect, inequitable, but functional — that has prevented the proliferation of nuclear weapons to dozens of nations that had the technical capacity to build them.

It took 23 years from Trinity to the NPT. During those 23 years, the world came within hours of nuclear war during the Cuban Missile Crisis, atmospheric testing contaminated the planet, and the populations of Hiroshima and Nagasaki provided the evidence of what the technology could do when used.

The governance arrived after two cities were destroyed and the world nearly followed.

The question for AI governance is whether the nuclear pattern can be compressed — whether governance can arrive before the equivalent of Hiroshima rather than after it.

The nuclear precedent provides both the model (international treaty, verification mechanisms, institutional enforcement) and the warning (the governance required catastrophe as a prerequisite).

The question is which aspect of the precedent will repeat.

> *"Nuclear governance took 23 years and required the destruction of two cities as a prerequisite. The question for AI is whether governance can arrive before the equivalent of Hiroshima rather than after."*

Why AI Is Harder

The nuclear parallel illuminates the governance challenge, but it also reveals critical differences that make AI governance harder than nuclear governance in several respects.

The technology is dual-use — capable of both civilian and military application — at every level. Nuclear weapons and nuclear power can be distinguished — imperfectly, but the distinction exists in the physical infrastructure required. Enrichment facilities produce observable signatures. Weapons-grade material has specific properties.

The IAEA can inspect. AI has no equivalent distinction.

The same model that writes a student's essay can generate disinformation.

The same training infrastructure that produces a medical diagnostic tool can produce a military targeting system.

The same code generation tool that helps Maya build a patient management system can help a malicious actor build malware.

There is no physical marker that distinguishes a beneficial AI system from a harmful one.

The capability is identical.

The intent is invisible.

The actors are not only states. Nuclear weapons are built by governments. AI is built by companies.

The NPT is a treaty between nations. A treaty between nations cannot directly govern companies whose operations span multiple jurisdictions and whose compliance depends on corporate willingness rather than sovereign obligation.

The Anthropic red lines, the OpenAI safety commitments, the Google DeepMind ethics board — these are voluntary corporate governance mechanisms.

They can be changed by a board vote.

They can be abandoned under competitive pressure.

They are not treaties.

Even where governance is not voluntary, it is contested. By mid-2026 the binding attempts had arrived — and with them the fight over whether they would survive. The European Union's AI Act set enforceable obligations for high-risk systems, though its own timetable was slipping: a proposed Digital Omnibus would push the core high-risk rules from August 2026 to late 2027, and the negotiation meant to confirm the delay collapsed that spring, leaving even the date in doubt. In the United States, more than a dozen states had written their own rules — California, Texas, Colorado, and others — a patchwork of binding law assembled in the absence of any federal standard. Then the federal government moved against the patchwork: a December 2025 executive order directed the Justice Department to challenge state AI laws in court and tied billions in broadband funding to their repeal, and in March 2026 the White House asked Congress to preempt them outright. Congress had refused before. The contest was unresolved.

This is keeper removal by other means. In the cockpit the keeper is bypassed by automation; in the legislature the keeper is dissolved by preemption. A law that compels disclosure, or an audit, or the right to challenge an output is an institutional keeper — a layer of judgment placed between a system and the people it decides for. Strike the law and the keeper is gone. Nothing crashes. No one dies. The removal looks like the ordinary friction of politics.

And the treaties that are being attempted have not closed the gap. For more than a decade, states have argued at the United Nations over autonomous weapons — systems that select and engage targets without a human deciding to fire. By 2026 the Secretary-General and the Red Cross had called on governments to conclude a binding instrument that year, and more than a hundred and twenty states had endorsed the goal. The instrument did not exist. The negotiations remained a rolling text, blocked for years by the handful of military powers building the weapons. The one domain where the actors are nations, where the stakes are human life, and where the principle — meaningful human control — is agreed by almost everyone, has still produced no treaty. If governance cannot bind the machine that chooses whom to kill, the prospects for binding the machines that choose whom to sentence, or hire, or surveil are not bright.

The proliferation is inherent. Nuclear nonproliferation works, imperfectly, because the materials and facilities required to build a nuclear weapon are difficult to acquire, expensive to maintain, and detectable by inspection. AI proliferation is inherent in the technology's nature. Open-source models can be downloaded by anyone.

The knowledge to train models is published in academic papers.

The hardware, while expensive for frontier models, is sufficient for capable smaller models at consumer prices. You cannot prevent the proliferation of AI the way you can prevent the proliferation of enriched uranium.

The knowledge is in the open.

The genie is out of the bottle.

The speed exceeds governance capacity.

The NPT took 23 years to negotiate. AI capabilities are advancing on a timeline measured in months. A governance framework negotiated through traditional diplomatic channels — the pace of the UN, the speed of treaty ratification, the tempo of multilateral negotiation — will be obsolete before it is signed.

The technology will have evolved past whatever the framework was designed to govern.

What Governance Would Require

Despite these difficulties, the nuclear precedent provides elements that any AI governance framework would need to include. A shared assessment of risk.

The NPT was possible because the major powers agreed on one thing: uncontrolled nuclear proliferation was a shared threat.

The Cuban Missile Crisis provided the catalytic moment — the event that made the risk undeniable. AI governance requires a similar shared assessment. Some elements of that assessment are emerging — the Bletchley Declaration of November 2023, signed by 28 countries including the US and China, acknowledged the risks of frontier AI.

But acknowledgment is not governance. A declaration is not a treaty. Verification mechanisms.

The International Atomic Energy Agency — the IAEA — provides the inspection and verification infrastructure for the NPT.

No equivalent institution exists for AI. Proposals for international AI safety institutes, model registries, and compute monitoring frameworks are under discussion but not yet operational at scale.

The technical challenge of verifying AI development is harder than verifying nuclear development: a data center training an AI model looks identical from the outside to a data center running a cloud computing workload. Institutional enforcement.

The IAEA has the authority to inspect facilities, report violations, and refer non-compliance to the UN Security Council.

No AI governance institution has equivalent authority.

The EU AI Act has enforcement mechanisms within Europe.

No institution has enforcement authority at the global level — the level at which the technology operates. A governance speed that matches the technology speed. This has no precedent.

Every previous governance framework was written for a technology that had stabilized.

The printing press did not evolve between Gutenberg and the Statute of Anne. Nuclear weapons advanced, but the fundamental physics was understood. AI is evolving faster than any previous governed technology. A framework designed for today’s AI may not address tomorrow’s.

The governance must be adaptive — designed not for a specific capability but for a class of capabilities that is changing as the rules are being written.

> *“Every previous governance framework was written for a technology that had stabilized. AI is evolving faster than any previous governed technology.*

The governance must be adaptive — designed not for a specific capability but for a class of capabilities that is changing while the rules are being written.”

Part III Restated

Part III has mapped the consolidation phase of AI’s five-phase cycle across two dimensions. Chapter 7 mapped the domestic consolidation: the concentration of AI infrastructure in a small number of companies, the vertical integration of the stack from chip to cloud to model to distribution, the orbital communications monopoly that wraps around everything, and the two forms of gatekeeping — infrastructure and capability — that are exercised by private actors without democratic mandate.

This chapter mapped the geopolitical consolidation: the competition between two superpowers with incompatible governance philosophies, the weaponization of the supply chain through export controls, the fragmented global regulatory landscape that cannot govern a planetary technology with national rules, and the nuclear parallel that provides both a model and a warning for what international AI governance might look like.

The consolidation is proceeding on both dimensions simultaneously. Domestically, a handful of companies control the infrastructure. Geopolitically, two nations are competing to control the capability.

The governance that could address either dimension — domestic regulation of essential AI infrastructure, or international coordination of AI development — does not yet exist in a form adequate to the challenge.

Part IV enters the room. Who is attempting to write the rules? What dilemmas do they face? Who is absent from the process?

And what happens when the three trajectories of this trilogy — power concentration, the removal of the keeper, and the absence of governance — converge?

* * *

CHAPTER GC-9

The Builders' Dilemma

On the companies that build the technology and the impossible choice they face between profit and principle

The Builders' Dilemma

In February 2026, Anthropic — the company that builds the Claude AI system — was offered a contract with the United States Department of Defense valued at approximately $200 million.

The company's leadership evaluated the contract and identified two applications that it determined were incompatible with its values: the use of its AI for autonomous weapons targeting and for mass surveillance of civilian populations. Anthropic declined those two applications while remaining open to other defense work.

The company did not refuse the relationship.

It drew lines within the relationship — two specific boundaries around the uses it would not permit.

The response from the defense establishment was swift and severe. Elements within the Department of Defense pushed to designate Anthropic as a supply-chain risk — a classification that would effectively blacklist the company from all government contracts, not just the disputed ones.

The argument was not that Anthropic's technology was inadequate.

The argument was that a company unwilling to provide unrestricted AI capabilities to the military could not be relied upon as a strategic partner.

The message was clear: if you build the technology, you do not get to decide how it is used.

The customer decides.

And this customer has a budget measured in hundreds of billions of dollars.

The public response was different.

Within weeks of the confrontation becoming public, Claude became the number one application in the App Store.

The company reported over one million daily sign-ups.

The growth was driven predominantly by new users — people who had never previously used the product, who chose Anthropic specifically because the company had drawn a line.

The market rewarded the refusal.

The structural tension that the Anthropic confrontation made visible — a tension that exists inside every company that builds frontier AI, and that no company can resolve alone.

> *"The defense establishment said: if you build the technology, you do not get to decide how it is used. The public said: we will choose the company that decides. Both responses were rational. They pointed in opposite directions."*

Five days before the Anthropic confrontation became public, an identical dynamic played out at a different scale. Canadian Prime Minister Mark Carney stood before the World Economic Forum in Davos and named the rupture in the rules-based international order — the speech documented in Chapter 8 of this book.

The response from the hegemon followed the same architecture as the response to Anthropic's red lines.

Carney was revoked from the Board of Peace. He was threatened with 100% tariffs. He was called "Governor Carney" — a deliberate diminishment of sovereignty.

The US Treasury Secretary described the oil-rich province of Alberta as "a natural partner for the US" while meeting with separatist groups seeking to fragment Canada from within.

And when Carney refused to retract, the administration claimed he was "aggressively walking back" his remarks — a claim Carney denied publicly and on the record: "I meant what I said in Davos." Anthropic drew a line and was threatened with blacklisting.

Carney named a reality and was threatened with economic devastation. The pattern is the same: the entity that self-governs — that exercises judgment about what is permissible — is punished by the entity with power.

The dilemma is not confined to companies.

It operates at every scale where an actor with less power attempts to constrain an actor with more.

The Dilemma Stated

The builders' dilemma is a coordination problem, and stating it precisely matters because the precision reveals why market solutions are insufficient. Scenario one: unilateral self-governance. A company governs itself — draws red lines, restricts certain uses, invests in safety research, declines contracts that cross its boundaries.

If its competitors do not do the same, the self-governing company loses market share, government contracts, and potentially its viability.

The customers who want the unrestricted capability go to the competitors who provide it.

The self-governance is punished by the market.

The company either abandons its lines or is outcompeted by companies that never drew them. Scenario two: universal self-governance.

Every company in the industry governs itself — adopts the same red lines, restricts the same uses, invests at the same level in safety.

No company loses competitive advantage because all companies bear the same constraints.

The problem is solved.

But this scenario requires simultaneous, voluntary, sustained coordination among competitors — companies that are, by definition, trying to outperform each other. Voluntary coordination of this kind is, in the language of game theory, an unstable equilibrium.

The incentive to defect — to quietly relax one's constraints while competitors maintain theirs — is persistent and structural.

One defection breaks the equilibrium.

And the competitive pressure is existential. Scenario three: no self-governance.

No company governs itself. All capabilities are deployed without restriction.

The chaos phase deepens.

The harms examined in Parts I and II of this book — and in the two books that precede it — accelerate. Eventually, the harms produce a crisis severe enough to trigger reactive government regulation. Reactive regulation, designed in crisis, is consistently worse than proactive governance: it is broader, blunter, less technically informed, and more likely to constrain beneficial uses along with harmful ones.

The Sarbanes-Oxley Act, written in the aftermath of the Enron scandal, imposed compliance costs on every public company to prevent the behavior of a few.

The Patriot Act, written in the aftermath of September 11, expanded surveillance authorities that have proven extremely difficult to roll back. Reactive regulation is the worst of all outcomes — and it is the outcome that The coercion extended beyond rhetoric.

In January and February 2026, the United States aggressively pursued the annexation of Greenland — an autonomous territory of Denmark, a founding NATO ally.

When European nations attempted to bolster Arctic security in response to what they interpreted as a direct territorial threat to a European ally, the administration threatened a 10% tariff on those nations.

The economic weapon was deployed not against an adversary but against allies who attempted to uphold the collective security framework that the United States itself had built.

The Greenland annexation threats, the Alberta separatist meetings, the tariff threats, the deliberate diminishment of an allied head of state — these are not isolated incidents.

They are the instruments of a power that has concluded that the rules-based order it built is now a constraint on its freedom of action.

The hegemon is not abandoning the room.

It is rebuilding the room with fewer seats, higher walls, and a lock on the inside.

The Board of Peace — established at the World Economic Forum in January 2026 to handle Gaza's postwar reconstruction, deliberately bypassing the UN Security Council, with the US as the only permanent UNSC member at the table — is the architectural drawing of the new room.

It is a governance structure designed by a hegemon to exclude the traditional multilateral checks and balances that the hegemon itself established eighty years ago at Bretton Woods.

> *"Self-governance alone is punished by the market. Coordinated self-governance is an unstable equilibrium. No self-governance produces reactive regulation — the worst outcome. The dilemma has no market solution. That is why it requires a room."*

The Landscape of Attempts

The Anthropic confrontation was the most visible instance of the dilemma, but it was not the first.

The AI industry's recent history is a catalog of attempts to self-govern — and the structural forces that undermine those attempts.

In 2018, Google established an AI ethics board — the Advanced Technology External Advisory Council.

It was disbanded within a week of its announcement, after controversy over the appointment of members whose views on climate change and LGBTQ rights were deemed incompatible with the board's mission.

The attempt to create an independent governance mechanism for Google's AI development collapsed before it held a single meeting.

In 2020, Google fired Timnit Gebru — the co-author of the Gender Shades research and one of the most prominent AI ethics researchers in the world — after a dispute over a paper that examined the risks of large language models.

The departure of Gebru, and the subsequent departure of her co-lead Margaret Mitchell, effectively dismantled Google's internal AI ethics team.

The message was legible: internal governance that conflicts with the company's product direction does not survive. OpenAI's evolution traced a different path to the same conclusion. Founded in 2015 as a nonprofit dedicated to ensuring that artificial general intelligence benefits humanity, OpenAI restructured in 2019 to a "capped-profit" model, then progressively moved toward a for-profit structure.

The original mission — to develop AGI safely and distribute its benefits broadly — was not abandoned in rhetoric.

But the structural incentives shifted: the company took billions in investment from Microsoft, became a commercial product company, and the safety mission was now competing for resources and attention with the revenue mission. Several senior safety researchers departed, citing concerns that safety work was being deprioritized relative to product development. Meta took yet another approach: open-source release of its Llama models.

The philosophy was that open access would democratize AI capability and prevent the concentration of power in a small number of closed-model companies.

The governance implication, however, was that open-sourced models are, by definition, ungoverned after release. Once the model weights are public, anyone can fine-tune the model for any purpose, remove safety guardrails, and deploy the result without the original developer's knowledge or consent. Meta's open-source strategy was a genuine contribution to the democratization of traced in The Last Keeper and AI capability.

It was also, simultaneously, a contribution to the governance vacuum.

This trilogy must be honest about the paradox. *Who Holds the Jar?* traces five thousand years of computational power concentrating in fewer and fewer hands and identifies that concentration as the existential danger. Open-source AI is the most powerful mechanism available for breaking that concentration — distributing capability to anyone who wants it.

This trilogy warns against concentration and warns against ungoverned distribution. Both warnings are correct.

And they pull in opposite directions.

The honest answer is that neither pure concentration nor pure distribution is safe. What is needed is distribution with standards — open access to capability, within frameworks that establish floors below which no deployment may fall. AI presents, and the fact that it is hard is not an argument for ignoring it.

The Structural Tension

The pattern across these cases is consistent and instructive.

Every major AI company has, at some point, attempted to govern its own technology. Google created an ethics board. OpenAI was founded as a nonprofit. Anthropic drew red lines. Meta open-sourced its models in the name of democratization.

And was constrained, compromised, or undermined by the same force: competitive pressure. Google's ethics board was disbanded because its existence created controversy that threatened the company's reputation. Google's ethics team was dismantled because its findings conflicted with product direction. OpenAI's nonprofit structure was abandoned because the capital requirements of frontier AI development exceeded what a nonprofit could raise. Meta's open-source releases advanced democratization but precluded downstream governance.

And Anthropic's red lines, however principled, exist at the pleasure of the company's leadership — they can be changed by a board vote, under competitive pressure, without external review.

The structural tension is this: building frontier AI requires billions of dollars in capital. Capital comes from investors who expect returns. Returns require revenue. Revenue requires customers. Customers include governments and enterprises that want unrestricted capability. Restricting capability restricts the customer base. Restricting the customer base restricts revenue. Restricting revenue threatens the capital.

And threatening the capital threatens the company's ability to build frontier AI at all.

The chain from safety commitment to existential risk is direct: a company that governs too aggressively may not survive to govern at all.

It is the explicit logic behind the supply-chain risk designation that was threatened against Anthropic.

The message from the defense establishment was: your self-governance is a risk to our supply chain. Your principled refusal is our strategic vulnerability. Govern yourself less, or be excluded from the market that funds your existence.

> *"The chain from safety commitment to existential risk is direct: a company that governs too aggressively may not survive to govern at all. The dilemma cannot be resolved from inside the market."*

Why Markets Cannot Solve It

The builders' dilemma is, in its formal structure, a multi-player prisoner's dilemma — a well-studied problem in game theory with a well-known result: in the absence of an enforcement mechanism external to the players, the equilibrium outcome is defection.

Every player is better off if all cooperate.

Every player has an individual incentive to defect.

The equilibrium is mutual defection.

The optimal outcome is cooperation.

The market produces the suboptimal outcome.

It is the insight that justified every governance framework described in Chapter 2.

The printing press required copyright law because publishers could not voluntarily coordinate to respect each other's rights. Electrical utilities required regulation because companies could not voluntarily coordinate to serve unprofitable rural areas. Nuclear weapons required a treaty because nations could not voluntarily coordinate to limit proliferation.

In each case, the coordination problem was solved by an external authority — a government, a regulator, an international institution — that set the rules and enforced compliance.

The AI builders' dilemma requires the same solution.

The companies cannot solve it alone. Voluntary self-governance is necessary but insufficient.

The solution is governance — rules set by an authority external to the market, binding on all participants, enforced impartially, and designed by a process that includes more than just the companies whose behavior is being governed. Not because companies are incapable of ethical behavior — some clearly are, and the Anthropic confrontation demonstrates it.

But because ethical behavior by individual companies, in a competitive market, is structurally unstable.

The room convenes not to replace the ethical actors but to protect them — to ensure that the company that draws a red line is not punished by the market for drawing it, because the red line is the law and the law applies to everyone.

The Anthropic Case Revisited

The Anthropic confrontation deserves a closer examination because it contains, in a single episode, every element of the dilemma and every element of a possible resolution.

The two red lines — no autonomous weapons, no mass surveillance — were not arbitrary.

They corresponded precisely to the two applications documented in *The Last Keeper* where the removal of the keeper is most consequential: AI targeting systems that compress human review to a formality (Chapter 9) and surveillance infrastructure that targets the keepers themselves (Chapter 10). Anthropic's red lines were, in the framework of this trilogy, a company saying: we have read the evidence, we understand the cost, and we will not build the tools that remove the keeper from these two loops.

The argument for the room. Not a new insight.

The defense establishment's response was also not arbitrary.

It reflected a genuine strategic concern: if the companies that build the best AI systems can unilaterally restrict their use by the military, then the military's access to AI capability depends on the ethical commitments of private companies — commitments that can change with a change of leadership, a change of market conditions, or a change of ownership. From the military's perspective, this is an unacceptable dependency.

And the dependency is real.

The public response — the million daily sign-ups, the number-one App Store ranking, the surge in users who chose Anthropic specifically because it drew the line

— provided a third data point.

The market, when given a choice between a company that self-governs and a company that does not, chose the one that self-governs.

The finding was not dispositive — consumer preference is not a governance framework — but it was significant.

It suggested that the market's response to self-governance may be more complex than the dilemma predicts.

The prisoner's dilemma assumes that defection is always rewarded.

The public response suggested that, in this case, cooperation was rewarded too. Whether this holds is an open question.

The Anthropic confrontation was a single event, in a specific political context, involving a company with a specific public profile and a specific user base.

It does not prove that self-governance is always rewarded by the market.

But it demonstrates something the dilemma's formal structure does not capture: the existence of a constituency — millions of users, representing billions of dollars in potential revenue — that values governance.

That constituency is a political fact.

It is the beginning of the demand side of the governance equation.

> *"The market, given a choice between a company that self-governs and one that does not, chose the one that self-governs. Consumer preference is not a governance framework. But a constituency that values governance is the beginning of the demand side of the equation."*

But this chapter must name a tension in its own argument.

The red lines this trilogy celebrates are an exercise of precisely the unilateral corporate power this trilogy warns about. Anthropic's refusal was not a democratic decision.

It was a corporate decision, made by a private company's leadership, affecting the military capabilities of a superpower, without legislative authority and without any mechanism that guarantees the red lines will hold beyond the current leadership's tenure. A different board, under different pressures, could draw the lines elsewhere or erase them entirely.

It is the argument for the governance framework that would make those red lines a matter of law rather than a matter of corporate courage — which is exactly what the next section proposes.

What the Builders Need

The resolution of the builders' dilemma is not more courageous companies.

It is a governance framework that makes courage unnecessary — that converts voluntary ethical commitments into mandatory minimum standards, so that the company that does the right thing is not penalized for doing it. Not an argument against the red The builders need a floor. Not a ceiling — not a regulatory framework that dictates every aspect of AI development and stifles innovation. A floor: a minimum standard below which no company is permitted to operate. A standard that covers the uses documented in this trilogy as most consequential: autonomous weapons, mass surveillance, deployment in healthcare without validation, deployment in criminal justice without transparency, deployment in critical infrastructure without independent audit.

A standard that applies to everyone — so that the company that self-governs above the floor is not competing against companies that operate below it.

The analogy is, again, to the governance of previous technologies. Pharmaceutical companies compete vigorously in the market.

They also all comply with FDA approval processes before their products reach patients.

The FDA is the floor — the minimum standard below which no company may operate. Above the floor, competition is fierce. Below the floor, competition is prohibited.

The floor does not eliminate innovation.

It channels innovation in directions that are compatible with the safety of the people the products affect.

The AI industry does not have a floor.

It has voluntary commitments from some companies, silence from others, and a competitive environment in which the absence of a floor rewards the companies that spend least on safety and deploy most aggressively.

The builders who want to self-govern are asking, in effect, for the room to convene and establish the floor that would protect their ability to compete responsibly.

The Dilemma as Invitation

The builders' dilemma is, paradoxically, the strongest argument for governance — because it comes from the builders themselves.

The companies that build frontier AI are not, for the most part, resisting governance. Many of them are requesting it.

They are requesting it because they understand the dilemma from the inside: self-governance is insufficient, competitive pressure is relentless, and the alternative to proactive governance is reactive regulation that will be worse for everyone.

The builders are not the only voices that need to be heard.

The next chapter asks who else should be in the room — and who is conspicuously absent.

The builders have resources, access, and technical expertise.

The people who bear the consequences of the builders' decisions — the patients, the defendants, the displaced workers, the surveilled communities, the nations whose sovereignty depends on infrastructure they do not control — have none of these things.

And the room, if it convenes, will produce a framework that reflects the interests of whoever is sitting at the table.

CHAPTER GC-10

The Missing Seats

On who is in the room, who is not, and what history teaches about whose voice shapes the rules

> *"The rules always reflect the interests of whoever is in the room. The question is never whether rules will be written. The question is: written by whom? And for whom?"*

The Missing Seats

Return, one last time, to Philadelphia.

The fifty-five delegates who gathered in the Pennsylvania State House in the summer of 1787 were, without exception, white men. Most were wealthy. Most were property owners. Many were slaveholders.

They were lawyers, merchants, planters, and military officers.

They represented twelve states — Rhode Island refused to participate — and within those states, they represented the interests of the propertied class that had selected them. Women were not in the room. Enslaved people were not in the room — though they were counted, at three-fifths, for the purpose of allocating the political power of the states that enslaved them. Indigenous nations, whose sovereignty the new government would systematically dismantle, were not in the room. Indentured servants, tenant farmers, the landless poor — the vast majority of the population — were not in the room.

The Constitution they produced reflected the interests of the people who wrote it.

It protected property rights.

It preserved slavery.

It restricted the franchise to propertied men.

It required two and a half centuries of amendment, litigation, civil war, and social movement to extend its protections toward — not to, but toward — the people who were not in the room when it was written. It demonstrates the oldest pattern in every governance framework in history: the rules serve the people who write them. Not exclusively. Not without possibility of amendment.

But initially, structurally, and persistently, the framework reflects the interests, the assumptions, and the blind spots of whoever is sitting at the table.

The question for AI governance is not only whether the room will convene.

It is who will be in the room when it does.

> *"The Constitution reflected the interests of the people who wrote it. It required two and a half centuries to extend its protections toward the people who were not in the room. The rules always serve whoever is sitting at the table."*

Who Is in the Room

The rooms where AI governance is currently being discussed — congressional hearing rooms in Washington, commission chambers in Brussels, summit venues in Bletchley Park and Seoul, boardrooms in San Francisco and Shenzhen — are populated by a specific and remarkably consistent set of actors.

The builders.

The CEOs and chief scientists of the companies that develop frontier AI systems testify before Congress, brief regulators, and participate in summits.

They bring deep technical expertise and The lesson of every governance significant institutional resources.

They also bring the structural interests of their companies: the desire for regulation that is predictable, that does not disadvantage them relative to competitors, and that does not constrain their ability to develop and deploy their products. Their presence is essential. Their interests are not the only ones at stake.

The lobbyists.

In 2023 and 2024, the technology industry spent hundreds of millions of dollars on lobbying related to AI regulation in the United States alone.

The lobbying operations employ former congressional staffers, former regulators, and former White House officials who understand the legislative process and have personal relationships with the people who will write the rules.

The asymmetry in resources is not subtle: the AI industry's lobbying budget exceeds the total annual budget of most civil society organizations that work on AI governance by orders of magnitude.

The regulators. Government officials tasked with understanding AI and crafting appropriate rules. Many are knowledgeable and committed. Many are also understaffed, under-resourced, and outmatched — attempting to regulate a technology that is evolving faster than their agencies can process, with budgets that are a fraction of the companies' R&D; spending, and with technical expertise that the private sector continually drains through higher salaries.

The revolving door between government and industry is well documented and structurally incentivized: the regulator who develops AI expertise becomes more valuable to the companies being regulated than to the agency doing the regulating.

The academics. Researchers from universities and think tanks who provide technical analysis, policy recommendations, and empirical research. Their contribution is invaluable but their influence is constrained by the same resource asymmetry: academic research operates on grant cycles and publication timelines that are measured in years, while the technology and the policy environment change in months.

Who Is Not in the Room

The people who bear the most direct consequences of AI deployment are, with few exceptions, not in the rooms where the rules are being discussed.

The patients. AI diagnostic systems are being deployed in healthcare settings that serve millions of people.

The algorithm systematically underserved Black patients by confusing healthcare spending with healthcare need.

The patients affected by that algorithm — and by the many similar systems deployed across healthcare — are not represented in AI governance discussions. Patient advocacy organizations exist but are not typically included in the technical and policy forums where AI healthcare regulation is shaped.

The clinical perspective is represented by healthcare institutions and professional associations, which are not the same as the patients themselves.

The defendants. AI risk assessment tools influence sentencing, bail, and parole decisions affecting millions of people in the criminal justice system.

The defendant Eric Loomis, sentenced to six years of initial confinement based in part on a proprietary risk score he could not examine, challenge, or understand — represents a category of affected person that has no voice in the governance process. Defendants do not testify at AI governance hearings. Incarcerated people do not submit comments to regulatory proceedings. Public defenders, who represent the population most directly affected by algorithmic sentencing, are among the most under-resourced institutions in the justice system.

The displaced workers.

The seven hundred Klarna employees from Chapter 6 — and the millions of workers across every white-collar profession who face displacement by AI systems — are not organized, are not represented by institutions with lobbying capacity comparable to the technology companies, and in many cases do not yet know that they are affected.

The displacement is distributed across industries, occupations, and geographies in a way that defies traditional labor organizing.

There is no union of workers displaced by AI.

There is no industry association that represents the economic interests of the people whose judgment is being automated.

The surveilled.

The communities monitored by the surveillance infrastructure — the Uyghur population in Xinjiang, the journalists targeted by Pegasus, the activists whose networks are mapped by intelligence agencies, the populations of the Global South whose faces train facial recognition systems they did not consent to — are not in the room.

They are, in many cases, in countries that have no effective voice in the international forums where AI governance is discussed.

The Global South.

The data that trains frontier AI models is drawn disproportionately from English-language internet content, but the models are deployed globally.

The content moderation labor that makes AI systems safe for Western users is performed disproportionately by workers in Kenya, the Philippines, and other low-wage countries.

The environmental cost of AI training — the energy consumption, the water usage, the carbon emissions — is borne disproportionately by communities near data centers, which are often located in regions chosen for cheap energy rather than environmental capacity.

The nations that contribute the data, the labor, and the environmental cost are not proportionally represented in the governance forums that determine how the technology is built, deployed, and regulated.

> *"The patients. The defendants. The displaced workers. The surveilled. The Global South. The people who bear the most direct consequences of AI deployment are, with few exceptions, not in the rooms where the rules are being discussed."*

The Asymmetry

The asymmetry between those in the room and those absent from it is not an oversight.

It is a structural feature of how governance is produced. Governance proceedings require resources: the time to monitor regulatory developments, the expertise to understand technical proposals, the staff to draft comments and testimony, the relationships to gain access to decision-makers, and the financial capacity to sustain engagement over the years that governance processes require.

The technology companies have these resources in abundance.

The affected populations, almost by definition, do not.

The asymmetry is compounded by a feature specific to AI: the affected populations often do not know they are affected.

The patient whose healthcare algorithm underestimates their need does not know the algorithm exists.

The defendant whose risk score influences their sentence does not know how the score was calculated.

The job applicant whose resume was filtered by an AI screening tool does not know the tool was used.

The invisibility of algorithmic decision-making — a theme that runs through every chapter of *The Last Keeper* — means that the people most affected by AI are often the people least equipped to advocate for their interests in the governance process, because they do not know their interests are at stake.

It is the governance problem, restated for the current era.

The people who bear the consequences of a technology's deployment have always been underrepresented in the governance of that technology.

The workers who died in factories were not consulted on factory safety regulations until labor movements forced their inclusion.

The communities downwind of nuclear test sites were not consulted on testing policy until decades of illness forced investigation.

The pattern is consistent: the affected population is included in governance only after the harm is documented, organized, and politically mobilized — a process that takes years or decades, during which the governance framework is shaped by whoever is already in the room.

The Proxy Problem

The absent parties are not entirely unrepresented. Civil society organizations, academic researchers, investigative journalists, and some government agencies advocate for the interests of the affected populations. These proxies perform essential work.

The ProPublica investigation that exposed COMPAS's racial bias.

The Obermeyer research that revealed the healthcare algorithm's discrimination.

The Citizen Lab investigations that documented Pegasus deployments.

The +972 Magazine reporting that described AI targeting systems.

Each of these was the work of proxies — researchers, journalists, and advocates who brought the affected populations' experience into the governance conversation.

But proxy representation has structural limitations.

The proxies are outspent.

The technology industry's lobbying budget dwarfs the combined budgets of every civil society organization working on AI governance.

The proxies are outnumbered: a congressional hearing on AI might feature four industry executives and one civil society representative.

The proxies are outpaced: the technology evolves faster than the research that documents its effects, which evolves faster than the advocacy that translates research into policy pressure, which evolves faster than the legislative process that converts pressure into law.

At every stage of the pipeline from harm to governance, the affected population is behind. Not an AI-specific problem.

And proxy representation carries a deeper problem: it is representation without consent.

The civil society organization that advocates for patients affected by AI healthcare bias was not elected by those patients.

The researcher who studies algorithmic sentencing was not appointed by the defendants whose liberty is at stake.

The proxies are self-selected advocates, often excellent, often effective, but not democratically accountable to the populations they represent.

The governance they produce may be better than governance produced without them.

It is not the same as governance produced by the people who bear the consequences.

> *"Proxy representation is essential and insufficient. The proxies are outspent, outnumbered, and outpaced. And they represent without consent — self-selected advocates, not elected by the populations whose interests they carry into the room."*

The Precedent for Inclusion

The history of governance expansion is, in every era, the history of missing seats being filled.

The labor movement filled the missing seats in industrial governance. Workers who had no voice in factory conditions organized, struck, and forced the creation of governance mechanisms — unions, labor boards, workplace safety regulations — that included their interests.

The process was violent, protracted, and only partially successful.

But it produced a governance framework in which the people who bore the consequences of industrialization had, for the first time, an institutional voice.

The civil rights movement filled the missing seats in political governance. Citizens who had been excluded from the franchise — by race, by gender, by economic status — organized, protested, litigated, and forced the extension of governance protections to populations the original framers had excluded.

The Fifteenth Amendment.

The Nineteenth Amendment.

The Voting Rights Act of 1965.

Each was the result of sustained political action by people who were not in the room demanding that the room include them.

The environmental movement filled the missing seats in industrial and energy governance. Communities affected by pollution, nuclear testing, and environmental degradation organized, sued, and forced the creation of environmental protection agencies, clean air and water regulations, and environmental impact assessment requirements.

The room expanded to include interests that the original occupants had not considered.

In each case, initially serves the people who convene it.

The excluded populations organize, mobilize, and force their way in.

The governance framework is amended to reflect a broader set of interests.

The process takes decades.

The harm during those decades falls disproportionately on the excluded.

The AI Challenge

The challenge for AI governance is that the traditional mechanisms for filling missing seats may not operate fast enough.

The labor movement took decades to produce governance frameworks for industrial labor.

The civil rights movement took a century to extend the franchise.

The environmental movement took decades to produce regulatory frameworks for pollution and environmental protection.

In each case, the governance expansion moved at the speed of political organizing — the speed at which affected populations can recognize their shared interest, form organizations, develop political capacity, and exert sustained pressure on governance institutions. AI deployment is moving at the speed of software releases.

The gap between the technology's impact and the affected population's capacity to organize a governance response is wider than at any point in the history of technology governance.

The patients affected by biased healthcare algorithms cannot organize a movement at the speed at which new algorithms are being deployed.

The workers displaced by AI cannot form a union at the speed at which jobs are being eliminated.

The communities subjected to AI surveillance cannot mount a legal challenge at the speed at which the surveillance infrastructure expands.

This means that the initial governance framework for AI — the rules written during the governance window that and closing — will almost certainly be written without adequate representation of the people most affected by the technology.

The seats will be missing.

The framework will reflect the interests of whoever is present: the builders, the lobbyists, the regulators who understand the technology, and the governments that view AI through the lens of strategic competition.

The question is whether this initial framework can be designed, from the outset, to include mechanisms for its own expansion — built-in pathways for the missing seats to be filled as the affected populations organize. A governance framework that acknowledges its own incompleteness and provides structural mechanisms for amendment is imperfect by design but improvable by structure. A framework that locks in the interests of its initial occupants and provides no pathway for expansion is not governance.

It is capture.

> *"A governance framework that acknowledges its own incompleteness and provides mechanisms for amendment is imperfect by design but improvable by structure. A framework that locks in the interests of its initial occupants is not governance. It is capture."*

The Cost of Exclusion

The cost of the missing seats is not abstract.

It has been documented, in specific cases, across two books and twenty-two chapters of this trilogy.

When the patient is not in the room, the algorithm confuses cost with need and denies care to the people who need it most.

When the defendant is not in the room, the algorithm encodes the biases of a discriminatory justice system and presents the result as objective.

There is one more missing seat that no governance framework has yet acknowledged, because the seat does not yet exist.

Every governance framework in this chapter — every term limit, every succession mechanism, every democratic rotation of power — was designed for holders who eventually die.

The biological substrate described in *Who Holds the Jar?*, Chapter 10, is building the technology to change that assumption. CRISPR, senolytics, telomere extension, DNA repair mechanisms borrowed from organisms that exhibit negligible senescence — these are funded, published, and in clinical trials.

If the holders of the jar disable mortality, the room's foundational assumption breaks.

No framework addresses the seat that never becomes vacant.

There is a category beyond the missing seat: the seat that does not exist. Not a chair left empty at the table, but a population for whom no chair was ever built.

In early January 2026, following a US military operation in Venezuela, the United States effectively severed Cuba's access to imported oil. Cuba had been under a US embargo for over six decades, but oil shipments — primarily from Venezuela and, intermittently, from Russia — had kept the island's infrastructure minimally functional.

When that supply line was cut, the consequences were immediate and catastrophic: nationwide blackouts lasting days, the collapse of hospital systems that depend on electrical power for ventilators, refrigeration of medicines, and surgical equipment, and severe food shortages as the cold chain for perishable goods disintegrated. Eleven million people experienced the downstream consequences of a geopolitical decision in which they had zero voice. Cuba does not attend the World Economic Forum. Cuba is not on the Board of Peace. Cuba is not a party to the semiconductor export controls, the AI governance summits, the quantum computing research consortia, or any of the frameworks where the rules of the computational era are being written.

Cuba is not at the table. Cuba is not on the menu. Cuba is in the kitchen — invisible, uncounted, and starving.

> *"There is a category beyond the missing seat: the seat that does not exist. Cuba is not at the table. Cuba is not on the menu. Cuba is in the kitchen — invisible, uncounted, and starving."*

The framework traced across this trilogy operates at every altitude.

In New York, you build the Oracle.

In Toronto, you negotiate its terms.

In Lusaka, you use it without knowing what is behind it.

In Havana, you lose your electricity because the Oracle's owners decided your oil supplier was an adversary, and no one in any room, at any altitude, considered your eleven million lives a variable worth including in the calculation.

The missing seats this chapter has described are seats that could, in principle, be filled.

The patient, the defendant, the worker, the surveilled community — each has advocates who could represent them, however imperfectly. Cuba represents something worse: a population whose absence from the governance structure is not an oversight.

It is a policy.

The embargo is the mechanism by which absence is enforced.

And the humanitarian cost of that absence — the blackouts, the collapsed hospitals, the hunger — is not a side effect.

It is the intended pressure.

The invisible population is invisible by design.

When the worker is not in the room, the displacement proceeds without safety nets, retraining frameworks, or transition support.

When the surveilled community is not in the room, the surveillance expands without constraint.

When the nation whose sovereignty depends on infrastructure it does not control is not in the room, the infrastructure operator makes decisions that determine the outcome of wars.

In January 2026, Canadian Prime Minister Mark Carney stood before the World Economic Forum in Davos and stated the principle that governs every missing seat this chapter has described: "If you're not at the table, you're on the menu." He was speaking about nations — about middle powers forced to choose between hegemons and hyperscalers.

But the principle applies at every scale.

The patient is on the menu.

The defendant is on the menu.

The displaced worker is on the menu.

The surveilled community is on the menu.

The nation whose sovereignty depends on a satellite constellation it does not own is on the menu.

The cost of the missing seat is not that your voice goes unheard.

It is that the rules are written to serve whoever is present — and you live under rules you had no part in writing.

The missing seats are not an abstraction to be addressed in a future amendment.

They are people whose lives are being shaped, right now, by systems they did not consent to, governed by rules they did not write, in a process they cannot access.

Every day that the seats remain empty is a day that the rules are written without their input and the consequences fall on them without their voice.

Part IV has now examined all the actors in the governance drama: the builders who face an impossible dilemma (Chapter 9), and the affected populations who are absent from the process (this chapter).

The next chapter is where the trilogy converges.

Three trajectories — the concentration of power, the removal of the keeper, and the absence of governance — are on a collision course. Chapter 11 maps the moment of convergence and asks what the intersection of all three looks like.

CHAPTER GC-11

The Convergence

On the moment when the trilogy's three trajectories meet, and the question that follows

> *"Three curves are converging. Power is concentrating exponentially. The keeper is being removed acceleratingly. And governance is proceeding linearly. The intersection is the present moment."*

The Convergence

This chapter takes its title from the book's title, and it delivers the thesis that the entire trilogy exists to state.

Three trajectories are converging.

They have been documented across three books, thirty-eight chapters, and a combined arc that stretches from a shepherd's counting jar in ancient Mesopotamia to an AI system that generates military targets in 2026.

Each trajectory has been traced independently.

This chapter maps their intersection.

The first trajectory is the concentration of power. Documented across fourteen chapters of *Who Holds the Jar?*, this trajectory shows that computational power has concentrated in fewer hands with every era — from the temple administrators who controlled the counting tokens, through the governments that controlled the mainframes, the companies that controlled the operating systems and processors, the platforms that controlled the internet, to the seven organizations that can train frontier AI models, the two companies that manufacture the chips, and the one company that controls sixty percent of active satellites.

The pattern is distribute-then-recapture: each era distributes capability and then reconcentrates control.

The concentration is accelerating.

The current era's reconcentration is happening simultaneously with its democratization, leaving no gap for governance to intervene.

The second trajectory is the removal of the keeper. Documented across twelve chapters of *The Last Keeper*, this trajectory shows that the human in the loop — the person who exercises judgment when the system encounters conditions it was not designed for — is being removed, degraded, overridden, deauthorized, outpaced, and overwhelmed in every domain where automated systems make consequential decisions.

The removal is justified each time by the same reasoning: the machine is faster, cheaper, more consistent.

The reasoning is correct on average.

The cost is paid at the margin — in lives, liberty, and justice.

The third trajectory is the absence of governance. Documented across the ten chapters of this book that precede this one, this trajectory shows that the governance frameworks that could arrest the first two trajectories — that could constrain the concentration and preserve the keeper — are fragmentary, inadequate, outpaced by the technology, captured by the builders, and missing the voices of the people most affected.

The room has not convened.

The rules have not been written.

And the window is closing.

> *"Power is concentrating exponentially. The keeper is being removed acceleratingly. Governance is proceeding linearly. The three curves are converging. The intersection is now."*

The First Convergence: Capability and Governance

AI capability and AI governance are on a collision course. Capability is advancing exponentially.

The performance of frontier AI models on standard benchmarks has improved at rates that consistently exceed predictions. Tasks that AI could not perform two years ago — writing competent code, passing professional licensing examinations, generating photorealistic images, conducting multi-step reasoning — are now routine.

The capability frontier moves forward in months.

The improvements are driven by increases in compute, data, and training methodology, all of which are following trajectories that show no sign of approaching a ceiling. Governance is proceeding linearly — through legislative processes, regulatory consultations, international negotiations, and institutional adaptations designed for a world that moved at a different speed.

The EU AI Act took approximately three years from proposal to enactment.

The US has not passed comprehensive AI legislation. International coordination on AI governance is in its earliest stages.

Each of these processes operates at the tempo of human institutions: hearings, comment periods, committee markups, floor votes, diplomatic negotiations, treaty ratifications.

The tempo is measured in years. An exponential curve and a linear curve will always intersect.

When they do, the exponential curve pulls away — the gap between capability and governance widens, and the governance becomes progressively less relevant to the technology it is attempting to govern.

The rules written for today's AI will not address next year's AI.

The framework designed for current capabilities will be obsolete by the time it is enacted.

The convergence of capability and governance is not a future event.

It is the present condition.

The Second Convergence: The Lego and the Lumber

Within the broader AI governance challenge, there is a specific convergence that affects every organization that builds or uses software — which is to say, every organization on earth.

The distinction introduced in Chapter 4 — between model-driven development (Lego bricks with built-in safety) and code-generated development (raw lumber without constraints) — is not an abstract architectural preference.

It is the front line of the governance battle, and it is being fought inside every enterprise, every government agency, and every institution that depends on software.

On one side: platforms that enforce structure. Enterprise software built on model-driven architectures that constrain what the builder can create to configurations the platform has validated.

The safety is in the system, not in the individual.

The platform catches the error the builder does not see.

The governance is embedded in the tool.

On the other side: AI tools that generate raw code from natural-language descriptions, with no platform enforcing structure, no automated auditing, no compliance verification, and no mechanism to catch the vulnerabilities that the builder cannot detect and the tool does not flag.

The capability is striking.

The governance is absent.

The convergence is this: both approaches are advancing simultaneously, and the organizations that must choose between them — or manage some combination of both — are making decisions that will determine the security, reliability, and auditability of the software that runs healthcare systems, financial markets, government services, and critical infrastructure for the next decade.

The choice between Lego and lumber is not a product decision.

It is a governance decision.

And it is being made, in most organizations, without a governance framework to guide it.

> *"The choice between Lego and lumber is not a product decision. It is a governance decision. It is being made in every organization on earth, without a governance framework to guide it."*

This convergence is a microcosm of the larger one. At the enterprise level, the question is: will we build software with systems that enforce safety or systems that leave safety to the builder? At the civilizational level, the question is the same: will we develop AI within frameworks that enforce accountability or in a vacuum that leaves accountability to the builder?

The answers at both scales will be determined by whoever writes the rules — or by the absence of rules, which produces its own answer.

The Third Convergence: The Trilogy

The three books of *The Cost of the Machine* converge in this chapter, and the convergence is not metaphorical.

It is visible in a single case study.

In September 2022 and February 2026, one company — SpaceX — demonstrated all three trajectories simultaneously. Power concentration (*Who Holds the Jar?*): Starlink controls over sixty percent of active satellites in low Earth orbit.

The capability to provide orbital internet connectivity is concentrated in one company to a degree that has no parallel in the history of communications infrastructure.

The distribute-then-recapture pattern: anyone can buy a terminal, one company owns the constellation.

The removal of the keeper (*The Last Keeper*): the decision to grant or deny connectivity — a decision that determines whether military operations succeed, whether hospitals stay online, whether a nation can defend itself — is made by one person, without independent review, without democratic mandate, without the governance framework that would place a keeper between the decision and its consequence.

The keeper in this case is not removed from the loop.

The keeper *is* the loop — one individual holding a switch that the governance system has not learned to govern.

The absence of governance (*The Great Convergence*): no treaty governs private orbital communications monopolies during armed conflict.

No international body has jurisdiction.

No governance framework requires the operator to maintain service, prohibits the operator from denying service, or provides a mechanism for review after either decision.

The room has not convened for this scenario because the scenario did not exist until one company built a constellation that two sides of a war both came to depend on.

One company.

One constellation.

One person's decisions.

Three trajectories — power, cost, and governance — intersecting in a case study that is not hypothetical, not projected, not modeled.

It has already happened.

The convergence is not an event to be anticipated.

It is a condition to be recognized.

> *"One company. One constellation. One person's decisions. Three trajectories — power, cost, and governance — intersecting in a case study that has already happened. The convergence is not an event to be anticipated. It is a condition to be recognized."*

A second convergence, at the level of nations, became visible within days of this book's writing.

When Canadian Prime Minister Mark Carney named the rupture in the rules-based order at Davos in January 2026, the response demonstrated all three trajectories simultaneously. Power concentration: the hegemon used trade threats, tariff escalation, and the deliberate encouragement of internal separatism to punish a sovereign nation for naming reality.

The removal of the keeper: the leader who exercised judgment — who said the thing the system needed said — was threatened with consequences designed to deter future truth-telling, not just by Canada but by every middle power watching.

The absence of governance: no international framework protects a nation from economic retaliation for speech at a diplomatic forum.

The rules that would have prevented the punishment do not exist — because the rules-based order whose protection they would have invoked is the very system Carney was declaring ruptured.

The Starlink case is a convergence of infrastructure.

The Carney case is a convergence of governance. Together they demonstrate that the three trajectories are not converging in a single domain.

They are converging everywhere — in technology, in diplomacy, in the relationship between nations and the infrastructure they depend on.

The convergence is systemic.

The Five Phases Applied to AI

The five-phase cycle traced across four technologies in Chapters 2 and 3 can now be applied to artificial intelligence with the specificity this chapter's position in the book demands.

Phase 1 — Innovation (2012–2020): The deep learning breakthroughs. AlexNet's image recognition in 2012. DeepMind's AlphaGo defeating Lee Sedol in 2016. GPT-2 demonstrating coherent text generation in 2019.

The capability existed.

It was expensive, concentrated in a handful of research labs, and inaccessible to the general population. Complete.

Phase 2 — Democratization (2022–present): ChatGPT, November 30, 2022.

One hundred million users in two months. Open-source models by 2023. AI code generation tools by 2024.

The Shopify mandate by 2025.

The capability distributed to hundreds of millions of people. Accelerating.

Phase 3 — Chaos (2024–present): Documented across Part II of this book. Ungoverned software creation at scale (Chapter 4).

The epistemic crisis of AI-generated disinformation (Chapter 5). White-collar labor displacement at unprecedented speed (Chapter 6).

The harms documented across *The Last Keeper* — biased algorithms, algorithmic amplification, AI targeting, mass surveillance — intensifying. Beginning.

Phase 4 — Consolidation (2023–present): Documented across Part III of this book. Seven frontier model builders.

Two chip companies.

Three cloud providers.

One orbital communications monopoly.

The vertical integration of the AI stack.

The geopolitical competition between two superpowers.

The distribute-then-recapture pattern operating in real time. Already visible.

Phase 5 — Governance: Fragments.

The EU AI Act. Executive orders. National strategies.

The Bletchley Declaration. Corporate self-governance that is structurally unstable.

No comprehensive international framework.

No Philadelphia.

No Bretton Woods.

No NPT. Incomplete.

The critical observation: Phases 3 and 4 are happening simultaneously. In the original cycle, democratization produced chaos, and the chaos produced consolidation, in sequence. AI is producing chaos and consolidation at the same time.

The builders who are consolidating control of the infrastructure are also the builders whose tools are producing the chaos of ungoverned deployment.

The entities that would need to be governed are the same entities that are consolidating the power that makes governance difficult.

This simultaneity has no precedent in the historical cases.

It means the governance window is not just short.

It is closing from both directions — compressed by the chaos that demands governance and by the consolidation that resists it.

“Phases 3 and 4 are happening simultaneously. Chaos and consolidation at the same time. The window is closing from both directions — compressed by the chaos that demands governance and by the consolidation that resists it.”

The Indicators

How would we know if the governance window is closing? What are the indicators that would signal the transition from a window that is open — where governance is still possible — to a window that has shut — where the consolidation is irreversible and the rules will be written by the consolidators?

The printing press provides the template.

The window closed when state censorship regimes became the primary mechanism of information control — when the consolidators (governments) wrote the rules (censorship) and the rules served the consolidators’ interests (suppression of dissent).

For AI, the indicators of a closing window would include:

The achievement of self-sufficiency in AI compute by a major power, eliminating the chokepoints that currently provide leverage.

The establishment of de facto technical standards by dominant AI companies that become impossible for regulators to override.

The embedding of AI systems so deeply in critical infrastructure that removing or modifying them becomes practically impossible.

The normalization of the historical record shows In prior technology cycles AI-driven decision-making in domains like healthcare, criminal justice, and military operations to the point where the demand for human oversight is treated as impractical rather than essential.

The political capture of regulatory institutions by the companies they are tasked with governing.

And the failure of international coordination — the moment when it becomes clear that national approaches have fragmented to the point where a coherent global framework is no longer achievable. Not all of these indicators have been triggered. Some are well advanced.

The embedding of AI in critical infrastructure is accelerating.

The normalization of AI-driven decision-making is proceeding in healthcare, criminal justice, and military operations. China’s drive toward compute self-sufficiency is underway.

The fragmentation of governance approaches is visible.

The window is not shut.

But it is closing, and several of the indicators that would signal its closure are trending in the wrong direction.

The Present Moment

We are, as of this writing in March 2026, in the governance window for artificial intelligence.

The window is open.

The question this chapter exists to answer is: what would it take to keep it open long enough for the room to convene?

The historical precedents suggest three requirements.

First: a shared recognition of urgency.

The NPT was made possible by the Cuban Missile Crisis — thirteen days that made the risk undeniable.

The Geneva Conventions were made possible by two world wars.

The printing press's governance arrived after 130 years of religious warfare. In every case, governance was catalyzed by a crisis that made the cost of inaction intolerable.

The question is whether AI governance can be catalyzed by the evidence already available — the costs documented in *The Last Keeper*, the concentration documented in *Who Holds the Jar?*, the chaos documented in this book — or whether it will require a crisis of a magnitude not yet seen.

The hope of this trilogy is that the evidence is sufficient.

The precedent is not encouraging.

Second: a coalition willing to act.

The NPT required the cooperation of the two superpowers — adversaries who agreed on one thing: uncontrolled proliferation was a shared threat. AI governance requires a coalition that includes the major AI-developing nations, the companies that build the technology, the civil society organizations that represent the affected populations, and the international institutions that can provide coordination and enforcement.

No such coalition currently exists. Elements of it are visible: the Bletchley Declaration, the bilateral discussions between the US and China on AI safety, the EU's regulatory leadership.

But a declaration is not a coalition. Discussions are not agreements. Regulation in one jurisdiction is not governance of a global technology.

Third: a framework designed for speed, not for a technology that had stabilized. AI has not stabilized and may not stabilize.

The governance framework must be adaptive — designed not for a specific capability but for a class of capabilities that evolves while the rules are being applied.

This has no precedent.

The closest analogue is financial regulation, which attempts to govern a system — global finance — that evolves continuously.

The track record of financial regulation is mixed: it has prevented some crises and failed to prevent others.

But it demonstrates that adaptive governance is at least conceptually possible, even for systems that outpace the regulators.

There is one domain where governance has already moved — quickly, and from an unexpected direction. The systems this book has described decide about people: their sentences, their loans, their targets. A different kind of system has begun to relate to them. Companion chatbots — built to be friends, confidants, partners — hold sustained, human-like conversations, and their use, including by minors, grew fast enough that by 2026 the harms had become concrete enough to legislate. The keeper question arrives here in its strangest form: when the system is experienced as a relationship, who watches it? The states answered first. New York's companion-AI law took effect in November 2025, California's in January 2026, with others following through the year. The laws are modest and revealing. They require the system to disclose, on a schedule, that it is not human; to remind minors to take breaks; and to maintain protocols that detect expressions of self-harm and refer the user to a crisis line. The keeper is being legislated back into the conversation — not as a person in the room but as a required behavior of the machine itself. It is a measure of how far the machine has come that the law must now insist a system tell a child it is not a friend.

The Moment

Power is concentrating.

The keeper is being removed. Governance is absent.

The three trajectories are not parallel — they are reinforcing.

The concentration of power makes governance harder, because the entities that would be governed have the resources to resist it.

The removal of the keeper makes governance more urgent, because the cost of ungoverned systems is measured in lives.

And the absence of governance accelerates both the concentration and the removal, because neither is constrained.

The reinforcing dynamic is the reason the convergence is dangerous.

Each trajectory amplifies the others. Concentrated power removes keepers faster (because the entities with power face less accountability). Removed keepers make concentration easier (because the humans who might challenge the concentration are no longer in the loop). Absent governance allows both to proceed unchecked.

The triangle tightens.

The final chapter makes the closing argument.

It returns to the room.

It names the choice.

And it asks whether the evidence documented across three books and thirty-eight chapters — five thousand years of computational history, a century of catastrophic failures, and the governance patterns of every transformative technology — is sufficient to convene the room before the window closes.

The reinforcing dynamic has a human measure.

In 1983, Stanislav Petrov saved the world because he had twenty minutes and the authority to override a machine.

The Soviet early-warning system reported five American nuclear missiles inbound. Petrov judged the data inconsistent with a genuine first strike and declared it a false alarm.

He was right — sunlight on high-altitude clouds had fooled the sensors.

The keeper worked because three conditions held: he had time, he had context, and he had authority.

The Lavender targeting system deployed forty years later gives its operator twenty seconds and a rubber stamp.

The governance question is not whether human judgment matters — Petrov proved it does.

The question is whether any framework will require it.

* * *

CHAPTER GC-12

The Choice

On September 17, 1787, the final day of the Constitutional Convention, and the question this trilogy leaves with the reader

The Choice

The choice is not between perfect governance and imperfect governance.

It is between imperfect governance and none.

The history documented across this trilogy provides no example of a transformative technology that was governed perfectly.

The Constitution sanctioned slavery.

The Geneva Conventions are violated in every war.

The NPT has not eliminated nuclear weapons.

The EU AI Act is a framework for one continent in a world where the technology is planetary.

Every governance framework in history has been imperfect, incomplete, shaped by the interests of whoever was in the room, and amended over time as the excluded populations forced their way to the table.

The history also provides no example of a transformative technology that was better off ungoverned.

The printing press, ungoverned for 130 years, produced the Wars of Religion. Electricity, ungoverned for decades, produced monopoly empires that collapsed on their customers.

The internet, never adequately governed, produced surveillance capitalism, platform monopolies, and a recommendation algorithm that played a determining role in genocide.

The absence of governance was not an absence.

It was a presence — the presence of the consolidators' rules, written in terms of service, in algorithmic design, in corporate policy, without democratic mandate and without accountability to the people who lived under them.

The choice for artificial intelligence is the same choice that every transformative technology has eventually presented. Not whether to govern — every powerful technology has been governed, sooner or later, well or badly.

But when.

And by whom.

And at what cost in the interim.

The Window

The governance window for AI is open.

The printing press took 350 years to govern.

The chaos phase killed millions. Electricity took 53 years.

The chaos phase produced monopoly abuse and financial collapse. Nuclear energy took 23 years.

The chaos phase nearly destroyed civilization.

The internet was never adequately governed.

The chaos phase is ongoing.

The Choice

> *"I confess that there are several parts of this constitution which I do not at present approve, but I am not sure I shall never approve them." — Benjamin Franklin, September 17, 1787*

On September 17, 1787, the final day of the Constitutional Convention, Benjamin Franklin rose to address the delegates.

He was eighty-one years old, the oldest person in the room, too frail to deliver his remarks himself.

He handed his written speech to James Wilson of Pennsylvania, who read it aloud. Franklin began by acknowledging that the document before them was imperfect.

He confessed that there were parts of it he did not approve.

He noted that he was not sure he would never approve them — a gentle admission that his own judgment was fallible and that time might reveal wisdom in provisions he currently doubted.

He observed that every delegate who came to the convention had brought their prejudices, their passions, their errors of opinion, and their local interests, and that the document reflected all of these as much as it reflected the delegates' collective wisdom.

And then he urged them to sign it. Not because it was perfect. Not because it represented the best governance framework that could theoretically be designed.

But because it was the best framework that this group of imperfect people, with their imperfect knowledge and their imperfect motives, could produce in this room, at this moment, under these constraints.

And because the alternative — no framework, no governance, no room — was worse. Franklin's speech was not a celebration.

It was a clear-eyed assessment of what governance actually looks like when human beings do it: compromised, incomplete, stained by the interests of whoever happens to be present, and incomparably better than nothing.

> *"The document was imperfect. Franklin urged them to sign it anyway — not because it was the best framework that could theoretically be designed, but because it was the best this group of imperfect people could produce at this moment. And because nothing was worse."*

The Trilogy's Argument

This is the final chapter of the final book of *The Cost of the Machine*.

The trilogy's argument can now be stated in full. *The Last Keeper* documented the cost of removing the human from the loop.

It traced the pattern from a lighthouse on a rock off Pembrokeshire through radiation therapy rooms, cockpits, courtrooms, clinics, social media platforms, military targeting systems, surveillance infrastructure, and software too complex for any human to verify. The keeper — the human who exercised judgment when the system encountered conditions it was not designed for — was removed, degraded, overridden, deauthorized, outpaced, or overwhelmed.

In every case, the removal was justified by the same familiar reasoning: the machine is better on average.

In every case, the cost was paid at the margin — in lives, in liberty, in justice.

The keeper's question: will there be a human watching the light when the mechanism fails?

Who Holds the Jar? documented the pattern that produces the cost.

It traced the concentration of computational power across five millennia — from the shepherd's counting jar through the abacus, the clockwork calculator, the mechanical engine, the electronic switch, the transistor, the microprocessor, the personal computer, the internet, and the neural network. In every era, the same pattern repeated: capability distributed, then control reconcentrated.

The distribute-then-recapture cycle produced a ratchet of power that tightened with each turn.

The jar's question: who will hold the computational power of the next era — and will it be possible to take it back? *The Great Convergence* documented the only force capable of arresting both patterns: governance.

It traced the five-phase cycle that every transformative technology has followed — Innovation, Democratization, Chaos, Consolidation, Governance — across the printing press, electricity, nuclear energy, and the internet.

It showed that the governance window is shrinking with each era, that the chaos phase of AI has begun, that the consolidation is already advanced, and that the room has not yet convened.

The convergence's question: who will write the rules — and what happens when nobody does?

Three books.

Three questions.

One argument: we build machines that accumulate power, we remove the humans who could check those machines, and nobody writes the rules to govern what we have built.

The cost is measured in lives, liberty, sovereignty, and the capacity of societies to govern themselves.

The pattern has repeated in every era.

The question is whether it repeats in this one.

> *"We build machines that accumulate power. We remove the humans who could check those machines. Nobody writes the rules. The cost is measured in lives. The pattern has repeated in every era. The question is whether it repeats in this one."*

The False Choices

The debate about AI governance, as conducted in the public sphere, is dominated by false choices that this trilogy's evidence should put to rest.

The false choice between innovation and governance.

The printing press was governed and continued to innovate. Electricity was governed and continued to power civilization. Nuclear energy was governed and continues to provide clean power and scientific capability. Aviation is governed by one of the most comprehensive regulatory frameworks in existence and is the safest form of transportation in human history.

In no case did governance eliminate innovation. It channeled innovation in directions compatible with human welfare.

The argument that AI governance will stifle innovation is not supported by a single historical example.

The argument that ungoverned AI will produce catastrophe is supported by every example The false choice between governance and speed.

The argument that governance is too slow for AI is an argument against bad governance, not against governance itself.

The NPT was negotiated in under Governance channeled documented across thirty-eight chapters.

The false three years once the political will existed. Aviation safety regulations are updated continuously as technology evolves. Financial regulation adapts to new instruments and new risks on an ongoing basis.

The governance framework for AI does not need to be a static document that takes a decade to write.

It needs to be an adaptive framework — a set of principles, institutions, and enforcement mechanisms that evolve with the technology.

The challenge is real.

The challenge is not a reason to abandon the attempt.

The false choice between national sovereignty and international coordination. AI governance does not require nations to cede sovereignty.

It requires nations to coordinate, as they have for nuclear weapons, for aviation safety, for financial stability, for climate change. Coordination is not surrender.

It is the recognition that a planetary technology requires a planetary response, and that national governance of a global technology is an oxymoron.

The false choice between perfect governance and no governance disguises inaction as prudence.

The argument goes: we do not yet know enough about AI to govern it well, so we should wait until we know more.

The printing press was not well understood when the Statute of Anne was passed. Nuclear physics was not fully understood when the NPT was signed. Governance is never written with complete knowledge.

It is written with the best knowledge available, and it is amended as knowledge improves.

The alternative to imperfect governance is not better governance later.

The alternative is the consolidation of power without constraint, the removal of the keeper without check, and the chaos phase without limit — the precise conditions documented across thirty-eight chapters.

The false choice between naming reality and surviving it.

In January 2026, Mark Carney stood before the most powerful economic forum in the world and said the thing that the room had been performing around for years: the rules-based order is ruptured, and the nations that pretend otherwise are accepting subordination while calling it sovereignty.

The cost was immediate and visible. Revocation from the Board of Peace. Threats of 100% tariffs.

The deliberate encouragement of provincial separatism by a foreign government.

The Treasury Secretary of the hegemon describing a Canadian province as "a natural partner for the US" while meeting with groups working to fragment the nation.

And the attempted rewriting of what was said — the claim that Carney was "aggressively walking back" remarks he publicly reaffirmed.

The cost of naming reality was real.

But what Carney gained was something the cost could not erase: a room. A standing ovation from world leaders.

The President of Finland calling it one of the best speeches heard at Davos.

The Secretary General of NATO saying "Canada is back." And most critically, a framework that other middle powers could organize around — the TPP-EU trade bridge, the G7 critical minerals buyer's clubs, the AI governance cooperation among like-minded democracies. Carney did not win the argument with the hegemon.

He did something more important: he convened the room that the hegemon's dominance had prevented. Imperfect. Incomplete. Exactly like Philadelphia.

And incomparably better than the alternative, which was silence.

The false choice says: name reality or survive.

The evidence says: naming reality is how the room convenes — because someone has to speak first, and the constituency that values truth does not know it exists until someone gives it a voice.

Philadelphia is the right model not because it produced perfect governance but because it produced *amendable* governance.

The Constitution that excluded most of humanity — no women, no enslaved people, no unpropertied men — also contained the mechanism for its own correction.

The amendments came.

They came slowly, at tremendous cost, and they are still incomplete.

But they came — because the document was written to allow revision.

The AI governance that this trilogy calls for must do the same: write the rules now, with the people available, and build in the amendment process that allows the missing seats to reshape the framework once they arrive.

What Governance Requires

The evidence documented across this trilogy points to a governance framework with five elements. A floor, not a ceiling. Minimum standards below which no AI developer may operate, covering the most consequential domains: autonomous weapons, mass surveillance, deployment in healthcare without validation, deployment in criminal justice without transparency, deployment in critical infrastructure without independent audit.

Above the floor, competition is unrestricted. Below the floor, competition is prohibited.

The floor protects the companies that self-govern from being undercut by those that do not. Keepers in the loop. Mandatory human oversight in domains where the cost of failure is measured in lives — the domains documented across *The Last Keeper*. Not human oversight as a checkbox that can be compressed to twenty seconds per target. Meaningful human oversight, with the time, the authority, the information, and the institutional support to exercise independent judgment.

The keeper must be present.

The keeper must be empowered.

The keeper must be practiced.

The governance must require all three. Transparency at the point of consequence.

The people affected by algorithmic decisions must be able to know that an algorithm was used, how it was used, and on what basis.

The defendant must be able to challenge the risk score.

The patient must be able to know that an algorithm influenced their treatment plan.

The job applicant must be able to know that an AI system screened their resume. Transparency does not require disclosing proprietary algorithms.

It requires disclosing that an algorithm was used, what its inputs were, and how the affected person can contest the outcome. International coordination. A framework that matches the technology's scope.

The AI equivalent of the IAEA — an international institution with the mandate to monitor frontier AI development, the technical capacity to evaluate risks, and the authority to coordinate national governance efforts.

The institution does not need to be perfect.

The IAEA is not perfect.

It needs to exist. Seats for the affected. Structural mechanisms that ensure the people who bear the consequences of AI deployment have a voice in its governance. Patient representatives in healthcare AI oversight. Defendant advocates in criminal justice AI policy. Worker representatives in AI labor transition planning. Civil society organizations from the Global South in international AI governance forums.

The seats will be imperfect.

The representation will be incomplete.

But the alternative — a framework written exclusively by the builders and the governments that fund them — will reflect exclusively their interests, as every The window for AI is measured in years, not decades.

The capability is advancing monthly.

The consolidation is already advanced.

The chaos is beginning.

The room has not yet convened in a form adequate to the challenge.

The evidence is sufficient.

Three books have documented the cost: lives lost to systems whose keepers were removed, power concentrated, and governance frameworks that arrived too late for the printing press, too late for electricity, too late for the internet, and nearly too late for nuclear weapons.

The evidence does not say that AI governance will succeed.

It says that the attempt must be made.

It says that the room must convene.

It says that imperfect rules, written now, by imperfect people, in an imperfect process, are incomparably better than the alternative.

And the alternative is not an absence.

It is a choice — made by default, by the people who benefit from the absence, at the expense of everyone else.

> *"The evidence does not say that AI governance will succeed. It says that the attempt must be made. It says that imperfect rules, written now, by imperfect people, in an imperfect process, are incomparably better than the alternative."*

The Jar. The Light. The Room.

Eight thousand years ago, a shepherd on the plains of Mesopotamia pressed a clay token into a jar.

The token represented a sheep.

The jar was the first ledger.

The ledger was the first computer.

And the question — who holds the jar? — has echoed across every era since, growing louder as the jar grew more powerful and the stakes of holding it grew higher.

Two hundred years ago, a keeper on a rock in the Atlantic kept a light burning through storms, beside the body of his dead companion, because the light was not for him.

The light was for the ships.

And the question — will there be a keeper watching the light when the mechanism fails? — has echoed through every automated system that removed the human who could catch the failure the machine could not see.

Two hundred and thirty-seven years ago, fifty-five men gathered in a room in Philadelphia, nailed the windows shut, argued for four months, and produced an imperfect document that has governed the most powerful nation on earth ever since.

And the question — who is in the room, and who writes the rules? — has echoed through every governance moment since, from Bretton Woods to Geneva to the NPT to the room that never convened for the internet.

Three images.

Three questions.

One argument.

The jar is filling with a power no previous era has contained.

The light is burning, but the keeper is being told the light does not need watching.

And the room — the room where the rules for this power and this light could be written — has not yet convened.

The window is open.

The jar is filling.

The keeper is watching.
The room is waiting. Who will be in the room? What rules will they write?
And will the rules arrive before the window closes?

> *"The choice is not between perfect governance and imperfect governance. It is between imperfect governance and none. And if the history documented teaches anything, it is that none is not an absence. It is a choice — made by default, by the people who benefit from the absence, at the expense of everyone else."*

The Cost of the Machine --- *Book One*

The Last Keeper

* * *

Book Two

Who Holds the Jar? *The Machine Evolves: A Natural History of Computation — and a Warning*

* * *

Book Three
The Great Convergence

> *Three books.

One argument.
The jar is filling.
The keeper is watching.
The room is waiting. *Pumulo Sikaneta* — March 2026*

* * *

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About the Author

Pumulo Sikaneta works in technology strategy, where he has spent years studying how organizations adopt, govern, and are transformed by the systems they build.

His professional work in partner enablement and solutions architecture has given him a close-up view of the gap between what technology promises and what it delivers — and of the humans who stand in that gap. The trilogy began with The Last Keeper and continued with Who Holds the Jar? — The Machine Evolves: A Natural History of Computation and a Warning. Together, the three books trace a single argument from the human cost of removing people from the loop through the five-thousand-year pattern of computational power concentration to the governance battle that will determine whether the pattern repeats or breaks.

The inspiration for this trilogy came, in part, from conversations with his brother, Dr. Tabo Sikaneta, a nephrologist at the Scarborough Health Network in Toronto and founding member of the African Caribbean Kidney Association. Dr. Sikaneta's two decades of clinical work documenting the disproportionate burden of kidney disease in immigrant and non-White communities — and his 2025 BMJ Open study demonstrating that these disparities persist even within universal healthcare — provided the human ground truth that animates this trilogy's argument: that the data our systems learn from carries the inequities of the world that produced it, and that no algorithm can be more just than the society whose data it consumes.

The author was born in Zambia, raised in Canada, and lives in the United States — three countries with three different relationships to the power structures this trilogy examines.

He believes that the most important skill in any field is the willingness to be corrected.

Curious by nature. Correctable by choice. Always learning. Often wrong. Let's discuss.